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Restoring Dunhuang Murals: Crafting Cultural Heritage Preservation Knowledge into Immersive Virtual Reality Experience Design

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ABSTRACT

With the development of digital technology, tourists have higher expectations for their experiences with digital cultural heritage (CH). However, there is limited research and design guidelines on how to effectively convey expert knowledge and transfer the value and connotations of digital CH to users in a virtual environment. Based on existing literature, we proposed a systematic design strategy for immersive virtual reality (IVR) systems, incorporating multimodal interaction, gamification, and storytelling for CH presentation, development, and promotion to the public. Accordingly, taking the virtual restoration of the Dunhuang murals as an example, we developed an IVR system that provided users with learning, interactive, and entertaining experiences, facilitating their transition from mere visitors to active learners while engaging with digital CH. We conducted a between-subjects user study involving 42 participants. The results demonstrated that our IVR system motivated users' interests in CH, promoted CH preservation awareness, and could be applied in the domain of digital CH experiences.

KEYWORDS

Immersive virtual reality;
cultural heritage;
multimodal interaction;
immersive experience;
Dunhuang mural
preservation

1. Introduction

Due to various factors such as the natural environment, historical influence, and tourist impact, cultural heritage (CH) sites are vulnerable to disease, degradation, and deterioration over time (Themistocleous et al., 2016). The preservation of CH is a common concern globally. In addition, how to facilitate the presentation, development, and promotion of CH to the general public is also a critical issue in the contemporary context. The public has limited opportunities to access physical conservation scenes of CH on-site to comprehend its cultural value and spiritual significance, due to constraints in field tourism, including time, cost, distance limitations, and other technical barriers. Moreover, the conventional learning methods for knowledge acquisition provided by existing traditional platforms (websites, films, textual presentations, etc.) have made it challenging to generate meaningful interactions and experiences for the public. However, the advantages of digital technology provide new avenues for the preservation, display, and distribution of CH (Bekele et al., 2018), which can enhance the cultural experience and the effectiveness of CH knowledge education (Dincelli & Yayla, 2022).

Researchers in the field of human-computer interaction are actively developing novel experiences that integrate digital technologies with CH. Current digital technologies

utilized in CH experiences include applications such as online 360-degree panorama CH tourism (Kenderdine, 2015; Ren & Chen, 2021), mobile augmented reality guides for CH (Galatis et al., 2016), interactive multimedia wearable systems (Brancati et al., 2017), etc. However, due to the complexity of CH, especially for the historical CH sites containing both tangible and intangible assets, the materiality of different technical mediums poses challenges in fully expressing CH attributes. In recent years, virtual reality (VR) technology has made significant advancements in terms of its usability, flexibility, and rendering capabilities (Plecher et al., 2019), effectively enhancing users' immersion. Consequently, an increasing number of researchers and designers have been working on applying VR technology to create virtual experiences for CH historical sites. Hu-Au and Lee (2017) emphasized the advantages of contemporary VR as a medium for establishing a connection between users and learning objects, thereby fostering empathy, comprehension, and creativity. VR has the potential to enhance digital CH experiences by cultivating empathy towards CH, deepening users' understanding of its significance, increasing learning engagement, and enabling authentic experiences. Although existing research cases allow CH to be accessed digitally when physical access is limited, they primarily focus on the 3D replication and reconstruction of the CH environment in VR (Antlej et al., 2018; Guidi et al., 2007;

Micoli et al., 2018). However, there is a lack of sufficient support for systematic design considerations that prioritize user interactive experiences in learning. In addition, the lack of contextual background and narrative knowledge results in fragmented knowledge for users, making it challenging for them to connect the CH to their own knowledge and affective experience. It is worthwhile for researchers and designers to explore how to make better use of immersive virtual reality (IVR) systems in interactive experiences to motivate users' interest in CH and facilitate CH preservation knowledge dissemination. Therefore, we formulated the following research questions:

RQ1: What are the IVR system design requirements for the presentation, development, and promotion of CH?

RQ2: What design strategies can provide users with a better systematic experience of the local history, sociocultural information, and natural environment of CH through IVR systems?

RQ3: How can the design strategies be followed and utilized in practical IVR system implementation?

In order to address these research questions, we first conducted a thorough literature review in [Section 2](#) to comprehend the existing studies and current limitations of IVR for supporting CH preservation, interaction in virtual restoration, and multimodal interaction, gamification, and storytelling in IVR systems to identify corresponding design requirements for supporting CH preservation. Then, in [Section 3](#), building upon the literature review, we proposed a design strategy that methodologically addresses user experience in a CH-oriented IVR system, including situated consistency, embodied motivation, and affective connectedness. In [Section 4](#), we described our design and development of a multimodal IVR prototyping system following the proposed design strategy for an exploration and virtual restoration experience in the Dunhuang Mogao Grottos, which focused on crafting preservation knowledge into the IVR experience. Our system was tested by 42 participants in a user evaluation study in [Section 5](#) using both quantitative and qualitative approaches to validate our design strategy for a CH-oriented IVR system. The results in [Section 6](#) demonstrated that our design strategy effectively stimulated users' interest in CH and enhanced their overall experience by fostering a deeper understanding of CH sites. In [Sections 7 and 8](#), we concluded this work and discussed its limitations and future work.

Our work makes two primary contributions, as follows: (1) We propose the design strategy for an optimized user experience in CH-oriented IVR systems and expand the research body around incorporating CH attributes into the fields of multimodal interaction, gamification, and storytelling. The strategy investigates the methodological approaches for engaging the general public in forming intuitive understanding of CH sites, embodied experience for CH-related activities, and preservation knowledge of CH connotations in order to support the purposes of CH presentation, development, and sustainable promotion respectively. (2) We present an IVR system that combines a roaming tour of the Dunhuang Mogao Grottoes with virtual mural restoration,

shedding light on the deeper characteristics of CH by combining local historical context, natural environment, and sociocultural information in a single entity rather than simulating only the physical entity of the CH site itself. Our insights not only provide a unique solution for experiencing Dunhuang murals but also give emerging support for how to improve immersiveness, attract users' active learning interest, and build emotional connection to cultural identity efficaciously through the development of public-oriented IVR design in the CH remote visiting experience.

2. Related work

2.1. IVR for supporting CH preservation

China, as an ancient country with a wealth of cultural relics, has been actively involved in the digital preservation, presentation, and promotion of CH in recent years. Wu (2019) emphasized that the accurate and complete presentation of the CH entity and its context, the full use of digital CH to generate deeper interaction between cultural content and the general public, and the sustainable development of digital CH are the three important goals in the application of digital technology to CH. Lively presentations and immersive experiences of tangible and intangible Chinese CH information facilitate the dissemination of the Chinese CH, which contributes to shaping the collective identity of Chinese cultural groups (Li et al., 2022).

VR systems have been largely applied in the CH sector and can be classified into three categories, according to the level of immersion and the technical capabilities, namely fully-, semi-, and non-immersive VR (Beck et al., 2019). Compared to non-immersive (e.g., desktop-based VR) and semi-immersive (e.g., CAVE system) VR systems, fully immersive VR systems isolate users visually from the real world by providing synthetic 3D environments with head-mounted displays (HMDs). Dincelli and Yayla (2022) identified five affordances for an IVR experience, namely embodiment, interactivity, navigability, sense-ability, and create-ability.

CH sites, especially historical sites, contain tangible assets such as physical entities and natural environments, and intangible materials such as sociocultural evidence, spatial atmosphere, etc. A holistic and interactive virtual environment is needed to display this diverse range of information effectively. An IVR system enables user-centered presentation and renders vulnerable historical sites digitally accessible (Bekele et al., 2018), allowing people to virtually visit and interactively explore the sites. Interactions in IVR increase users' perceptual and cognitive levels, encourage them to capture the spatial-temporal relationships and meanings of cultural objects (Ferdani & Pietroni, 2021), and support their activities in addition to observations that are practically difficult or unfeasible in reality (Slater & Sanchez-Vives, 2016). This means that IVR is an extremely promising tool for the enhancement of cultural learning and sustainable tourism.

2.2. Interaction in virtual restoration for dunhuang murals

As one of the first historical sites in China that was recognized as a World Cultural Heritage site by UNESCO in 1987 (Li et al., 2022), Dunhuang Mogao Grottoes have always been at the center of attention in the fields of Chinese CH preservation and promotion. Dunhuang murals have been facing gradual degradation due to historical, natural, or manmade reasons. Experts are exploring and implementing virtual restoration methods to digitally preserve and reconstruct the murals for their integrity (Fu et al., 2017; Xia et al., 2018). However, it is difficult for the public to intuitively understand the current fragile situation of the murals, the restoration efforts, and the achievements of mural preservation experts, which hinders them from identifying and contextualizing related cultural knowledge.

Virtual restoration techniques for CH are usually developed for archaeological or academic purposes, or as visualization approaches for public exhibitions with limited interaction. In recent years, the focus on the public-oriented immersive and interactive experience of Dunhuang murals based on virtual restoration has been increasing. Current applications mainly involve the utilization of existing restored images or 3D models to construct an IVR environment, where users make limited movement and browse additional information using handles (Jin et al., 2021). More complex interactive elements such as multimodal environments, simulated activities, narratives metaphors, and emotional impact (Chu & Mazalek, 2019; Pietroni & Adami, 2014) are propitious to make the public's virtual experience a communicative and cognitive unity, aiming to reenact the sociocultural context of the CH, as well as making scientific process explicit to the public (Ferdani & Pietroni, 2021).

As the previous work of this article, Fu et al. (2020) firstly proposed to invite users to conduct active learning by simulating the physical restoration based on the virtual restoration results of the murals in Dunhuang Cave 320. Comparing with conventional virtual restoration displays, the embodied knowledge and situated experience incorporated in the IVR systems can be more efficacious in attracting interest, popularizing knowledge and cultivating deeper awareness of the preservation of Dunhuang murals.

2.3. Multimodal interaction in IVR systems

The subjective correlation of immersion is presence, which requires the system to allow users to perceive through natural sensorimotor contingencies (Slater & Sanchez-Vives, 2016). In the majority of immersive VR systems, users are usually overstimulated by the visual (sight) and auditory (hearing) modalities (Dincelli & Yayla, 2022). Displaying and manipulating other senses, such as haptic (touch) or olfactory (smell) sensory stimuli, is important in enhancing naturalness and immersiveness in a virtual environment, especially in cultural venues such as galleries, libraries, archives, and museums (Konstantakis & Caridakis, 2020). Multimodal experiences increase the users' sense of physical

presence, so that they become more involved in IVR to foster a deeper understanding of CH knowledge.

The haptic sense is unique for its proximity, strong immediacy, and ability as a two-way communication channel (Hayward et al., 2004). Olfaction also acquires information at close range and has not yet been explored in IVR systems to the same extent as the other dominant senses (de Paiva Guimarães et al., 2022). Users' haptic and olfactory experiences of the IVR systems affect the way they approach and process virtual objects and can be simulated via various stand-alone or wearable display devices. The majority of haptic interactions in CH mainly focus on the haptic reproduction of CH artifacts in the users' hands, using desktop-based force feedback devices (Ceccacci et al., 2021; Dima et al., 2014), or haptic gloves (Chrysanthakopoulou et al., 2021).

In addition to physical objects, the environments of historical sites also contain various haptic and olfactory features that provide rich information concerning the materiality of the CH sites, such as geological and climatic conditions, local vegetation, building materials, etc. Wind feedback is often used in IVR systems to simulate weather changes. Haptic Around (Han et al., 2018) utilized a fan, a hot air blower, mist creators, and heat lamps to provide hybrid-haptic sensations in a 2 m by 2 m space. Season Traveller (Ranasinghe et al., 2018) integrated a fan, Peltier elements, and micro air-pumps into the HMD, aiming to simulate real-world conditions such as scents, ambient temperatures, and wind patterns. Olfactory displays can also help users locate or identify certain objects in the virtual environment when moving around. Herrera and McMahan (2014) developed a desktop-based olfactory device to simulate the odor of designated objects when users were nearby. de Paiva Guimarães et al. (2022) developed a non-intrusive, low-cost, wearable olfactory display for HMD integration to provide olfactory cues of objects when users moved through the virtual environment. However, few of these applications have been incorporated into the CH sector.

2.4. Gamification and storytelling in IVR systems

Many CH-related applications allow users gradual exploration and discovery through observation, with limited interaction with the virtual environment. The lack of gamified mechanisms and storytelling makes it more difficult for IVR systems to capture the users' interest and attention, which leads to a decreasing desire to continue exploring and learning (Ibrahim & Ali, 2018). The gamification of CH applications can convert the communication of CH knowledge into an entertaining experience (Konstantakis et al., 2019); benefit from "the power of games to immerse, engage and motivate, and the capabilities of games to foster and facilitate cognitive gain, awareness, and behavioral change" (Liarokapis et al., 2017); enhance the overall IVR experience; and reach a wider audience (Cecotti, 2022).

To explore historical sites in depth, the audience often has to learn knowledge that is distant from their time. Storytelling, as a relatively perceptual method of knowledge transfer, can facilitate users' reception and understanding of

the diverse information of CH sites in a non-linear way (Li & Xu, 2021), while activating their emotional engagement (Trichopoulos et al., 2022). The integration of storytelling in gaming and its application in the cultural sector have become important forms of contemporary cultural education (Mariotti, 2022). Furthermore, the introduction of virtual characters in the virtual narrative system helps users acquire information naturally in their visual and voice interactions, transforming them from mere observers to active participants in the virtual environment (Ma et al., 2019).

Based on a narrative plot line, the virtual tour system of the Beijing-Hangzhou Grand Canal invited users to test their mastery of the relevant knowledge by introducing the concept of game-based “experiential learning” (Pan et al., 2020). In the VR application of the Sanjiangkou virtual CH site, four design strategies were adopted in the narrative process, including narrative attraction, user reflection on content, the connection between users and virtual scenes, and detailed design in VR applications, with the aim of bringing users into the perspective of storytellers and encouraging them to interact with the different scenes (Leow & Ch’ng, 2021).

Researchers have already attempted to incorporate gamification and storytelling into the design of relevant IVR systems for Dunhuang murals. Han et al. (2020) developed an IVR system that allowed users to take a flying carpet to change their location and viewpoint in the caves, even in mid-air, and to examine repaired sculptures closely with virtual flashlights. Li et al. (2019) applied animated Feitian figures and digital mural comic strips to the IVR system of the caves and utilized digital characters as narrative guides. Users could also activate the slow growth of a bodhi seed through meditation. However, current applications still have limitations in terms of the diversity of scenarios, interactivity, and communication of mural restoration knowledge.

3. Design strategy for CH-oriented IVR system

In this study, as a preliminary investigation, we attempt to provide users with an immersive, comprehensive, and engaging digital experience of CH through IVR systems, facilitating their exploration, understanding, and emotional connection with CH. The main tasks of digital experiences for CH involve three stages: CH presentation, development, and promotion. To fulfill the design requirements of multimodal interaction, gamification, and storytelling, our study proposes a design strategy for the IVR system in each stage, incorporating three main design considerations and six key factors (Figure 1): (1) Multimodal interaction for CH presentation: Constructing a multimodal virtual environment forms the foundation for creating an immersive presence and establishing an intuitive connection between users and CH sites. Situated consistency, comprising situated sense-making and situated metaphor, is the primary design consideration during this stage. (2) Gamification for CH Development: Gamifying characters, stories, and tasks motivates users’ embodied interaction and actively engages them in learning activities within the CH context. The main design consideration in this stage is embodied motivation,

including behavioral motivation and role motivation. (3) Storytelling for CH promotion: Affective connectedness, as the main consideration at this stage, including emotional reflection and empathic imagination, contributes to the sustainable promotion of knowledge transmission and the awareness of CH preservation. Storytelling helps bridge users’ psychological distance from CH and encourages their emotional experience of its cultural and spiritual significance.

The design strategy proposed in this study is an iterative process, and each of the above stages is interconnected in a sustainable sequence. Researchers and designers can adapt the design methods and guidelines according to the different stages and emphasis of the digital experience of cultural heritage. This approach facilitates the creation of a systematic digital CH experience.

3.1. Multimodal interaction for CH presentation: Situated consistency

Situated consistency refers to the establishment of a coherent perceptual connection between the virtual environment of an IVR system and the physical assets of historical CH sites. Dervin’s sense-making methodology (SMM) mainly emphasizes the contextualized sense-making processes (Reinhard & Dervin, 2012), which are driven by the user’s sensory and cognitive factors and have a direct impact on the acceptance and adoption results of the user experience in an IVR system.

The situated sense-making factor focuses on the initial situational impressions created by the IVR environment on the user’s sensory modalities. It involves leveraging multimodal interactions to arouse the human sensory system, create a perceptual context, and achieve proprioceptive wholeness. For instance, in the Dunhuang context, the use of slopes and distant buildings with high and low undulations can create a spacious and borderless outdoor natural space, while closed visual spaces, auditory echoes, and strong olfactory layers can be used to immerse the user in the interior environment of the Dunhuang caves.

While traditional VR systems emphasize the subjective “gaze” experience, IVR systems with multiple scenarios may result in fragmented, discontinuous, and disconnected visual fields. To address this, metaphorical assimilation effects can be employed to compensate for the transience of IVR scenarios and enhance the real bodily experience between users and the virtual environment (Gibbs et al., 2004). For example, the ambient wind causes a change in body perceived temperature from hot to cold and creates an atmosphere that changes from the outdoors to quiet and solemn indoor scenes. The combination of perceptual and metaphorical contexts forms a continuum within the IVR system, enabling users to progress towards the goals of the CH-based IVR system. It helps users understand how to interact with the IVR system environment, develop initial understanding and perceptions, and foster positive attitudes towards the tangible and intangible features of the CH site.

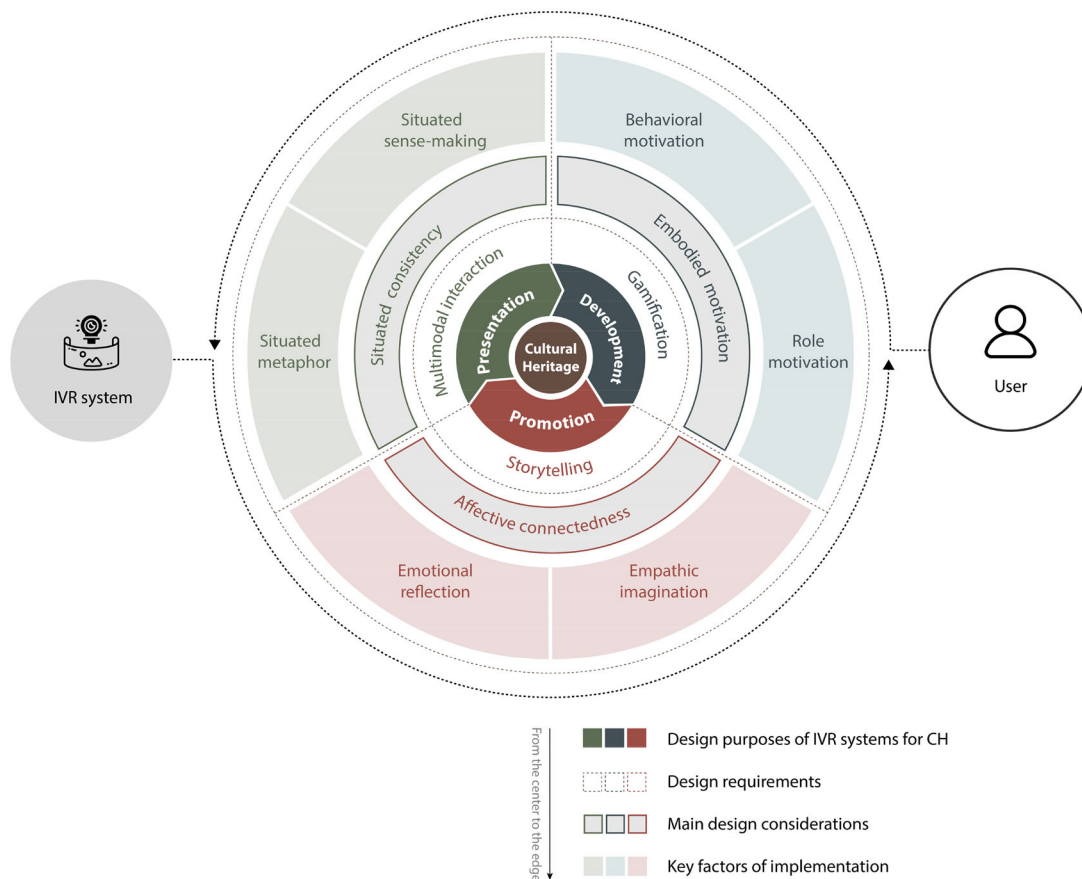


Figure 1. The design strategy for the user experience in the CH-based IVR system.

3.2. Gamification for CH development: Embodied motivation

Embodied motivation refers to the interaction design of the CH-related artifacts in an IVR system that empower user behavior and identity perception to promote users' actions. It aims to promote user actions, functionality, task completion, and sustained interest in exploring the IVR system. Embodiment is a crucial aspect of bodily interaction for engaging experiences (Gibbs et al., 2004). Behavioral motivation refers to interactive behavior, meaning that users are bodily engaged with artifacts through environmental design, level setting, and reward mechanisms in gamification, as well as learning and mastering CH-related expertise by utilizing devices to accomplish tasks, such as the physical restoration of Dunhuang murals. Technology plays a contextual role in the supporting devices (van den Eede, 2015), and the design of artifacts and techniques in IVR systems should prioritize accessibility, usability, and ease of use to aid users' comprehension of the technological and cultural symbolism of the IVR systems.

In addition, embodied cognition emphasizes the anchoring of the user's role perception during bodily-engaged activities. The perspective users adopt when entering gamified IVR scenarios directly impacts their cognitive identity and emotional connection with the IVR environment. The integration of additional character settings, collaborative devices, and environmental constructions in the IVR system should be carefully considered to foster role motivation.

Particularly for tasks with a unique cultural context in CH, role motivation should include appropriate cultural background support and ability compensation to boost users' self-efficacy in performing tasks and enhance their sense of immersion in the designated role to improve user retention time.

3.3. Storytelling for CH promotion: Affective connectedness

Situational consistency and embodied motivation within the IVR system lead to fostering affective connectedness between users and the CH sites presented. This process involves resonating with the spiritual content of the CH and influencing users' sustained attachment to CH promotion. The engagement mechanism of storytelling narratives in the IVR system facilitates the harmonization of users' mind and body.

According to Trauer and Ryan (2005), "place" is the center where emotional interaction occurs in the tourism experience. Emotional reflection, influenced by users' cultural background and prior experiences, plays a pivotal role in creating place-mediated memories in the digital experience, thereby fostering a user-centered perspective. Despite users experiencing CH sites through IVR systems in a distant location, emotional reflection establishes a meaningful social connection between users and CH sites (Lee & Shen, 2013), thereby rendering inaccessible heritage sites approachable.

Empathic imagination is another key factor for fostering an affective connectedness between users and the “others” (non-player characters, local artifacts, etc.; Low & Altman, 1992), to shift the users’ awareness towards CH preservation. Empathic imagination provides users with an intimate communicative space for “dialogue” through co-participation within the IVR system, such as a dialogue between users and CH across time and space, and to co-create with other users who participate in the IVR experience at different times. For sustainable CH development and promotion, it is important to stimulate users’ empathy for CH preservation and their perceptions of future digital CH presentations. Affective connectedness encourages users to transcend mere spectatorship and develop a sense of ownership toward CH, while perceiving CH as the precious cultural wealth of a nation and to further participate in the promotion of CH preservation.

3.4. Practice of the design strategy

We took Dunhuang Mogao Grottos as a representative case in the following sections to further practice our design strategy outlined above. Following the methodological design considerations, we developed a multimodal IVR prototype system to engage users in exploring Dunhuang Mogao Grottos and experiencing the virtual restoration of the murals. The entire IVR experience was not only designed to present the murals themselves, but also to establish a cohesive experience by leveraging the localization characteristics of Dunhuang mural cultures as a cue.

We intended to create a cultural sphere related to the Dunhuang murals by incorporating social elements, historical narratives, and the natural environment into the IVR system design as a whole, so that users could gain a deeper understanding of the background knowledge in a relatively short amount of experience time. To achieve this, we implemented multimodal interactions to simulate the local environment of Dunhuang in a way that actively engages users’ senses and encourages instinctual interaction with virtual objects. Simultaneously, gamified passages were implemented to encourage users to maintain a constant level of interest and focus while completing the exploration and learning tasks.

In addition, our system attempted to introduce multi-character narratives in an effort to strengthen their emotional connection with CH. We hoped that incorporating the aforementioned design strategies into the IVR system would broaden the experience spectrum for remotely visiting CH sites in the museum scenario. We aimed to construct a more active and sustainable awareness of CH and foster the development of cultural identity. The following section introduces the design concept and implementation of our IVR system for Dunhuang Mogao Grottos.

4. IVR system design and development

4.1. Design decisions and design materials

To facilitate an embodied learning experience in the virtual environment, we developed a three-phase progressive

narrative structure that guides users from being mere visitors to becoming skilled apprentices, gaining knowledge about mural restoration along the way. We designed three distinct simulated environments that correspond to different phases of tasks: (1) The desert scenario, representing the geographical environment of Dunhuang Mogao Grottos located in the north-west of China; (2) The gallery scenario, simulating the experience of the real visiting route of the Nine-Storey Pagoda in Dunhuang; (3) The cave scenario, recreating the exact environment of Cave 159 in Dunhuang Mogao Grottos, to present a fully immersive and enclosed space dedicated to the mural restoration task.

In the desert scenario, the primary focus was to cultivate an understanding of the geodesic layers and their component materials, which serve as the fundamental elements of the Dunhuang murals. Users were assigned the task of learning about the geodesic layers and collecting materials in situ to begin the journey. To enable users to comprehend the relationship between mural materials and the local vegetation of Dunhuang, we devised a fully immersive desert environment that included a 3D representation of the iconic Crescent Lake. This environment aimed to recreate the natural habitat where these materials are found. By incorporating simple interactive elements, users were able to collect materials within the desert scene, allowing them to gain preliminary insights into the mural culture, local natural scenery, and indigenous craftsmanship before entering the Nine-Storey Pagoda.

In the gallery scenario, we incorporated official video and textual content sourced from the Dunhuang Academy to provide users with a comprehensive introduction to the background of Dunhuang murals, mural diseases, and restoration techniques. To provide an accurate virtual restoration experience, we investigated the six main mural diseases (mold, discoloration, smudging, blistering, flaking, and exfoliation). We selected mold, discoloration, and exfoliation as virtual restoration targets. As these key forms of the physical restoration process were relatively easy to simulate, users would be able to clearly compare the mural before and after restoration. In addition to these three types of damage, the key steps of dusting before restoring any mural and smoothing the pigment layer were added, as they are crucial steps in physical restoration and prerequisites for all restoration work.

In the cave scenario, To enhance the authenticity of the cave environment, we utilized high-resolution fresco images of Cave 159, captured on-site at the Dunhuang Mogao Grottos, to generate textures for the cave scenes, enabling users to navigate the walls realistically. In addition, the five steps were paired with 3D restoration tools, such as dusting balls, scrapers, brushes, medicine levels, and cotton swabs, similar to what was used in the actual physical restoration process to enhance the user experience in terms of use. During the restoration simulation process, we used the digital restoration result layers achieved in the previous study (Fu et al., 2020) to guide the user through the restoration of the Bodhisattva mural in Dunhuang and present the user with the virtually restored effect of the statue as feedback at the end of the restoration experience.

4.2. Storyboard and interaction design

The virtual interactions in this study were character-based gamified stories to integrate narrative content into the restoration experience and provide role motivation for users. Specifically, the mural character “Xiaohua” was used to communicate with users and was involved in a plot that enabled the user to learn about restoration knowledge, understand the steps, and move the story process forward. In addition, we utilized a segmented apprentice-style narrative structure (Hay & Barab, 2001), based on the design of three distinct scenarios transitioning from collecting materials in the desert environment to the exploration of the indoor gallery

scene and finally to the restoration experience in the cave. This sequential narrative design allowed users to gradually acquire knowledge about murals and restoration, develop relevant skills, and employ the gathered materials in subsequent scenes (Figure 2).

4.2.1. Chapter 1: Preparing materials for mural restoration in the desert

The initial scene was based on the desert scenario with Crescent Lake, a landscape of sand and water symbiosis that simulated an immersive desert environment that transported the visitor to a geographic landscape unique to the area

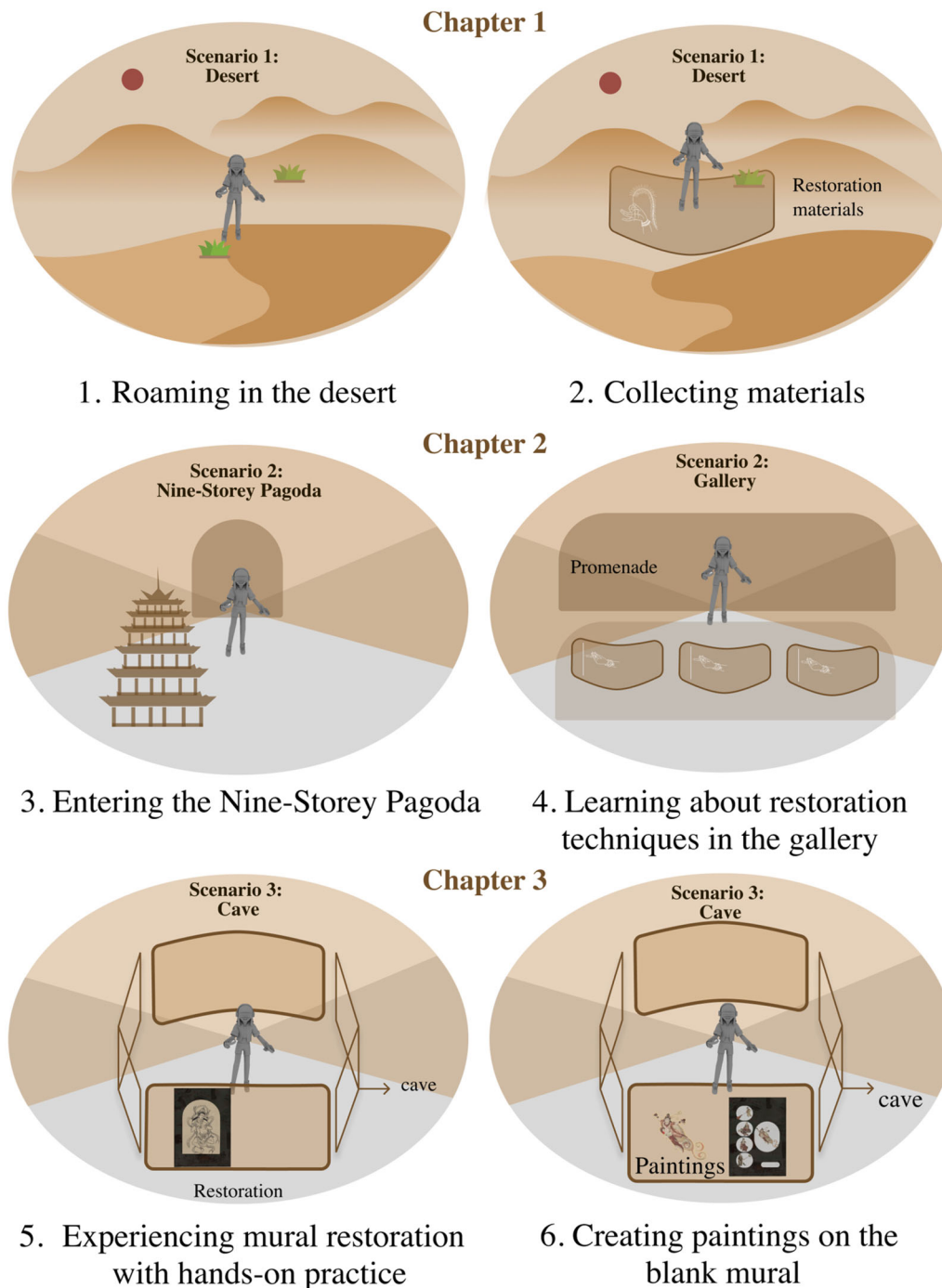


Figure 2. Storyboard of three chapters.

where the mural originated (Figure 3(a)). A 2D agent of the mural figure “Xiaohua” provided annotations including audio and visual guidance, and, after a short introduction, the user selected the button “start your journey” to enter the collection process. Upon clicking the button, a 3D deer, symbolizing the story of the Nine-Colored Deer depicted in the mural, would appear. Users would then follow the deer’s path to various nodes, where they could collect essential materials such as wheatgrass, flax, sand, and clay. The guidance of characters facilitated users’ roaming within the desert and the completion of material collection tasks, which aligned with the design strategies of situated sense-making and behavioral motivation. Each collection triggered corresponding introductory texts, enhancing the interactive experience and providing users with further insights into the mural’s historical background. Once all the materials were gathered, the deer would guide the user to a mobile location, initiating the transition to the next scene.

4.2.2. Chapter 2: Accumulating knowledge for restoration techniques in the gallery

The cave romp served as the second chapter of the experience (Figure 3(b)). Following the trail of the Nine-Colored Deer, the appearance of the Nine-Storey Pagoda and the monolith in the twilight provided a staged narrative experience. Once inside the Nine-Storey Pagoda, the user would arrive at a gallery that simulates the rock texture of the cave and contains paintings with signs of deterioration to introduce visitors to restoration in a step-by-step manner. The gallery featured two introductory videos: Dunhuang murals, mural diseases, and four disease restoration steps: dust removal and cavitation, mold removal, and coloring. Illustrations in the style of the Dunhuang murals were also used to remind users of the operational techniques with tools for each disease. These correspond to the subsequent mural restoration sessions, and after the user had viewed all the introductory videos, they were rewarded with the appropriate restoration tools as they learned to repair the murals. Once the restoration of all three diseases had been activated, the user could view the kit and explore the cave.

4.2.3. Chapter 3: Experiencing mural restoration with hands-on practice in the cave

The mural restoration was the final chapter, and we restored Cave 159 in a 1:1 proportion to create a fully immersive experience. As mentioned in Section 4.1.2, there were murals from Cave 159 on all four walls and the ceiling of the cave. The murals on one wall had four cue points that the user could interact with via flashing lights to learn the story and history of the murals (Figure 3(c)). After the user entered the cave at the entrance, the user’s perspective would be shown with the prompt screen and the Buddha that needed to be repaired. A dialog box first prompted the user to pick up the tools (e.g., how to use the handle buttons, etc.), and then the user followed the voice prompts of “Xiaohua” to complete the four processes of dusting, leveling, mold removal, and coloring. After completing these

steps, the user would hear an introduction to the current restoration techniques and a presentation of the results. We prepared a free creation session demo at the end of the restoration experience, and, once the restoration was complete, the user could pick up a paintbrush and enjoy creating on the blank wall of the mural or step on prepared stones in order to work towards the higher parts of the wall.

4.3. Multi-modal interaction design

4.3.1. Sound design

The background music for the desert scenes was composed using the Yayue Heptachord, a traditional Chinese tuning system that was commonly used in ancient rituals and ceremonies in Tang destiny. It aimed to set the desired emotional ambiance in the aural dimension, evincing the religious and philosophical mood associated with Dunhuang culture. The orchestration was carefully selected to preserve a sense of solemnity, which was achieved by reducing low frequencies. Complementary to these adjustments, the glockenspiel and harp, utilizing broken chords, were employed to evoke lively and engaging sound interaction experiences that brought musical aesthetic enjoyment while users completed targeted tasks.

The background music orchestration of the restoration process was based on percussion to cultivate a serene spatial atmosphere, allowing users to concentrate on the mural restoration. The sound effect was created with 44.1 kHz/24-bit high-precision sound samples for mixing. Adjustments were made to the volume, and specific frequency bands were attenuated using the 3D spatialization function of the Wwise middleware. This facilitated real-time modifications at interactive points, ensuring synchronization between the sound effects and environmental sounds as users moved within the virtual environment. The sound design enriches the situated auditory sense-making of the IVR environment, aiming to enhance the participant’s immersion and presence in the digital experiences of Dunhuang Mogao Grottos.

4.3.2. Olfactory design

In this study, we incorporated olfaction and wind feedback to provide invisible information as situated metaphors within the IVR environment. These elements served as task annotations, offering spatial hints and aiding users in developing an understanding of the IVR environment. To achieve this, we designed the settings of wind-driven olfactory devices that simulated scents and air currents in various IVR environments. By leveraging changes in scent and airflow, we aimed to create situated metaphors that facilitated the transition between indoor and outdoor scenes. The shifting direction of the wind also served as a cue, guiding users towards the target interaction points.

Presenting odor stimuli in VR applications offers the potential to introduce spatiotemporally controlled olfactory experiences to the virtual environment (Simon et al., 2023). Therefore, we built a prototype system using a simple air delivery unit as a starting point (Figure 4). This prototype

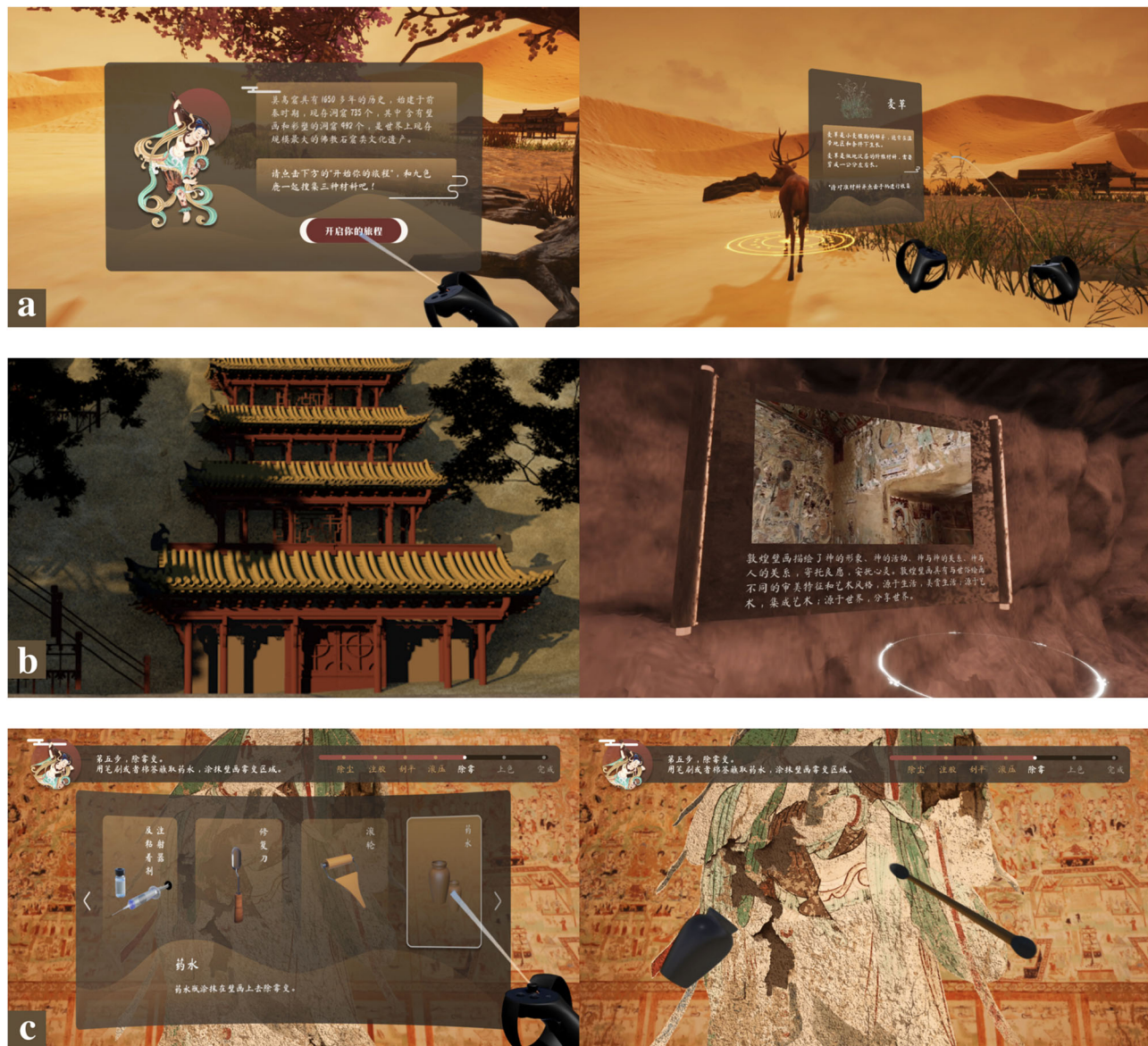


Figure 3. Interface design in the IVR system (a) Roaming in the desert and collecting materials (b) Entering the nine-story building and articulating knowledge of restoration techniques in the gallery (c) Selecting restoration tools and repairing the murals.

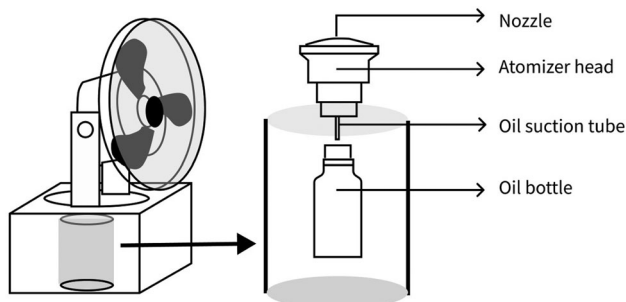


Figure 4. Olfactory device design.

contained a storage bottle for mixed odors, an odor atomizer, and an air pump for delivery. With this system, we could implement the concept that the user, when approaching the interaction point, would feel the smell of air coming from the device near the physical experience location. A total of four identical air transfer devices were prepared and

placed in the front, back, left, and right directions to be used in combination with the tactile wind described below, as shown in Figure 5. Scent and wind were turned on and off simultaneously during the experiment to explore the possibility of providing interactive cues through olfaction and somatosensory sensation.

4.3.3. Wind feedback design

In addition, we added haptic exposure in the form of airflow to the device for a more realistic experience. On the one hand, when the VR interface was limited to audio and visual channels, the user could not feel the physical (mechanical) feedback of the objects in the virtual world. So, the user experience was similar to being blindfolded and deaf. Therefore, haptic feedback has been used in recent years in virtual environments to draw users' attention to specific points of interest (Lindeman & Yanagida, 2003), provide collision information and haptic sensations (Sallnäs et al.,

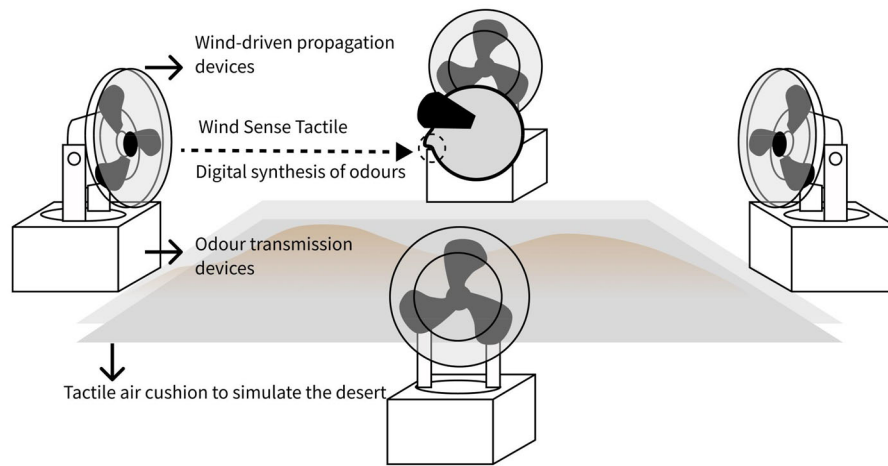


Figure 5. The overall design of the device.

2000), and improve spatial awareness overall (Komerska & Ware, 2003; Lécuyer et al., 2003).

On the other hand, considering that the user was moving during the IVR experience, it was difficult to reach the user with the above prototype scent transmission at close range. We borrowed the method of using vortex rings to generate olfactory fields by breaking the vortex ring containing the scent source (Nakaizumi et al., 2006) with an air cannon (Watkins, 2002) so that the scent can spread out. We combined these two points by assembling the olfactory device and the tactile device so that the user would feel the wind and smell the transmitted scent when the airflow blew on their face. We used four 40 W mechanical fans to provide the air supply in the physical environment. Four hybrid devices (containing smell and touch) were placed at the midpoint of the rectangular edge bounded by an air cushion, at the center of which was the location of the user experience (Figure 5). When the user switched from one of the three scenes to the next, the olfactory and haptic interactions with enhanced embodied learning would appear naturally.

4.3.4. Tactile air cushion design

Walking in the desert differed from walking in corridors and caves. The tactile feedback from walking in the desert was soft. It had been proven that tactile and pneumatic devices could act on feet to simulate terrain changes (Kobayashi et al., 2018; Schmidt et al., 2015). However, current research is focusing on shoes, so there are many inconveniences in the experience process. We wanted to have a more convenient pneumatic interface to simulate the tactile feedback of the desert. Therefore, we chose two different air cushions to simulate the ground in the experimental environment. The first one was a $173 \times 173 \times 27$ mm board floating on the sea, which was quite thick. Even if it was only filled with one-third of the air, it still fluctuated significantly when one walked on it. The second one was a $210 \times 140 \times 12$ mm inflatable mattress, with little fluctuation when stepping on it. The two main researchers walked on the air cushion with VR equipment, and, after repeated comparisons, the air cushion with less fluctuation and more desert-like feedback

was chosen (Figure 6). When the user left the desert scene and entered the Nine-Storey Pagoda, the system would control the air pump to slowly extract the air from the air cushion so as to simulate the ground transition from the desert to the corridor.

4.4. System implementation

The system ran on a PC with a 2.1 GHz Intel Core i7 12700 F CPU, an NVIDIA GeForce RTX 3060Ti 8 G GPU, and 16 G RAM. We chose to use the Pico Neo 3 for the IVR system, which included a head-mounted display device and two wireless handheld controllers, and we used Sony WH-1000XM5 noise-canceling headphones connected via Bluetooth to the Pico Neo 3's audio system to shield the environment from the noise generated by the fans and the deflation. The IVR application was developed with Unreal Engine version 4.27, which supported lossless import of 3D models from C4D, high-quality 3D rendering, dynamic simulation of natural environmental light sources, and a flexible Blueprint Visual Scripting system. The spatial scene was the basis of the IVR system. After importing the C4D model, we adjusted and processed the 3D model through some basic options of the rendering engine, such as meshes, materials, shapes, and behavior trees, so that the visual effects of the spatial environment and objects were consistent and harmonious. Simultaneously, we designed the guidance path and interactive behaviors of the Nine-Colored Deer through the behavior tree, such as turning its head, eating grass, roaming, etc., so that it could naturally lead the user to complete the task of collecting materials in the desert scene. In addition, we used Real-Time Render Target Painting to achieve mural restorations and support the user in freely creating visual effects on the broken mural.

The multimodal systems were all controlled by an Arduino Uno and communicated with the PC through the serial port. Unreal Engine communicated with the Arduino through the OSC protocol to operate the wind-activated tactile device, olfactory device, and air pump. The air mattress was inflated with a 600 W pump, and there was a ball valve to connect the servo motor (Teng et al., 2019) to

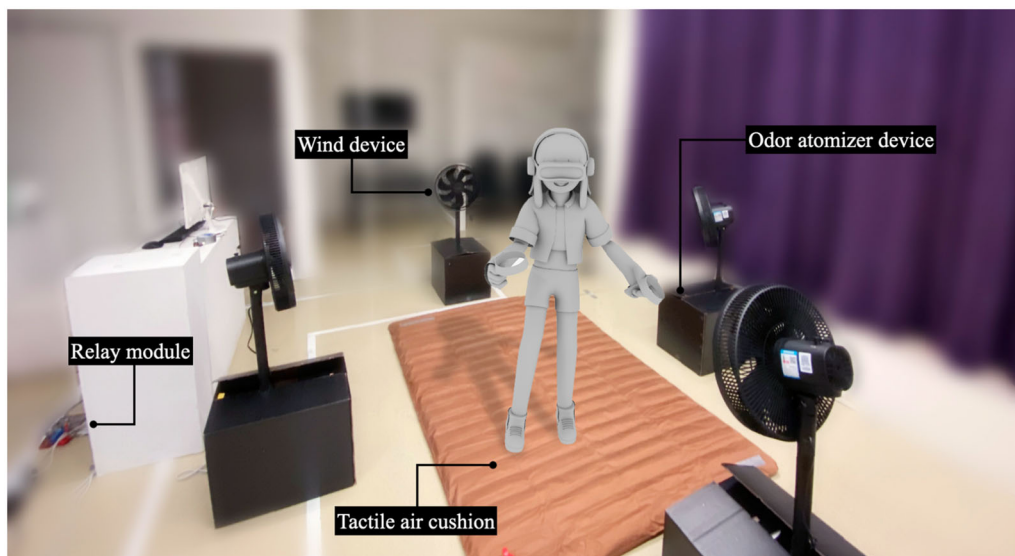


Figure 6. The experimental site and devices.

prevent air pressure leaking from both sides, while allowing a more convenient connection with the IVR system. Due to the loudness of the air pump, we used a 30-meter air hose to move it to another room to minimize the noise.

5. User evaluation

5.1. Participants

We conducted a between-subjects user study with 42 participants (P1-P42) to compare our multimodal IVR interactive experience system with a conventional VR system. The participants experienced the restoration of the Dunhuang murals in the IVR environment. The experimental group used our multimodal system, which provided tactile sensations, wind, and fragrances, while the control group experienced the same VR scenery and tasks without multimodal elements. Following the IVR experience, semi-structured interviews were conducted with participants from both groups to gain insights into their digital experiences and perceptions of the restoration process of the Dunhuang murals. By comparing the responses and feedback from the two groups, we aimed to assess the impact and effectiveness of our multimodal IVR system in providing an enhanced and more engaging experience for users.

By publishing posters on social media platforms, we were able to recruit a total of 42 participants, 14 males and 28 females. 74% of the participants were between the ages of 18 and 25, and 17% were between 26 and 30. They were randomly divided into two groups, with 21 participants using the multimodal IVR system (P1-P21) and the other 21 (P22-P42) in the control group (no multimodal VR system). More than 70% of participants had used VR devices before and were aware of the various effects of wearing a head-mounted display. Before formally engaging in the experiment, they all understood the goal of the study and provided written consent to participate.

5.2. Study procedure

This study was conducted in a controlled university laboratory environment characterized by a quiet setting. Prior to the commencement of the VR experience, participants were required to complete a pre-test questionnaire encompassing demographic information, their previous experience with virtual reality, their understanding and interest in cultural heritage, and their general VR familiarity. The purpose of the pre-test questionnaire was to gather initial self-reported data, aiding researchers in exploring the relationship between participants' experiences in IVR and their prior backgrounds. Further information regarding the questionnaire can be found in [Appendix A](#). Subsequently, participants were guided to the testing area, where they received instructions on operating the VR system and the buttons on the controllers. Following a familiarization session with the system's functionality, participants embarked on the IVR experience, which lasted approximately 15–20 minutes.

Participants were asked to fill out two rating-scale questionnaires to evaluate the system after finishing the IVR experience. At present, researchers have affirmed the importance of user experience in CH applications and proposed relevant evaluation dimensions (Konstantakis et al., 2020). The evaluation of a CH-oriented IVR system usually involves more aspects, such as assessing the cognitive load of participants through the NASA-TLX scale (Skola et al., 2020); measuring the presence, immersion, and flow in the experience process through the Immersive Virtual Environment (IVE) scale (Tcha-Tokey et al., 2017); and evaluating the perspicuity, dependability, and attractiveness of the system through the User Experience Questionnaire (UEQ) (Konstantakis et al., 2022). We finally selected the IVE scale questionnaire and the NASA-TLX workload assessment questionnaire, which are often used to evaluate user experience in VR environments. The IVE questionnaire was a compilation of several well-known questionnaires, and it surveyed the user experience in eight sub-scales: presence, engagement, immersion, flow, emotion, judgment,

experience consequence (VR-related sickness), and technology adoption (focused on the use of controllers) (Tcha-Tokey et al., 2016). On a 10-point Likert scale, responses ranged from one (“not at all”) to 10 (“completely agree”). The questionnaire was only available in French and English, so we translated it into Chinese. The NASA-TLX was answered on a 21-point scale, with subscales including mental demand, physical demand, temporal demand, performance, effort, and frustration (Hart & Staveland, 1988). Participants responded according to the actual situation during the participation process.

Subsequently, semi-structured interviews were conducted with the participants using an interview outline that encompassed the following aspects: (1) immersion within the IVR environment; (2) interactive experience and engagement in usability and task completion; (3) Knowledge acquisition of CH; and (4) Perceived sensory experience. Each interview session lasted approximately 40 minutes, and participants were remunerated with an approximate payment of 40 RMB, commensurate with their participation time.

5.3. Data analysis

We used a combination of quantitative and qualitative methods to analyze the data from the questionnaires and the semi-structured interviews respectively. The Shapiro-Wilk test was used to analyze the normal distribution of the IVE and NASA-TLX scales. The mean and standard deviation were used as descriptive statistics, and the Welch’s paired sample parametric T-test was performed to establish the significance of the experience difference between the multimodal group and the control group. The NASA-TLX scale did not conform to the normal distribution; therefore, the median, upper, and lower quartiles were used as

statistics, and the Mann-Whitney U test was applied to the nonparametric analysis.

The qualitative interviews were recorded with the expressed consent of all the participants. We used the Feishu Miaoji software to transcribe these audio files from speech to text. Three researchers simultaneously proofread the contents of these transcribed texts. Two researchers then conducted open coding on the interview text files from the perspectives of the overall experience, narrative, gamification, knowledge acquisition, and multimodal experience. A combination of qualitative and quantitative research approaches enabled a deeper understanding of the user experience. We elaborated on the abovementioned topics in the results and discussion section.

6. Evaluation results and discussion

More than 85% of the participants in this study were students, and more than 90% have a bachelor’s degree or higher. More than 70% of them had used VR equipment previously (Figure 7(a)), primarily for playing video games, watching films and television shows, and artistic creativity (Figure 7(b)). They primarily used VR devices at home and in shopping malls; only eight users (27%) had used VR devices in museums, art galleries, or other public spaces. The VR content they had experienced before included cultural artifact introduction, storytelling, CH interaction, and experience. More than 86% of participants in previous experiences had encountered situations in which they did not know how to proceed and were unable to access useful hints. In addition, they encountered issues such as being unable to detect their spatial location and becoming lost in the VR environment; experiencing vertigo and discomfort; having to deal with positioning and identification errors,

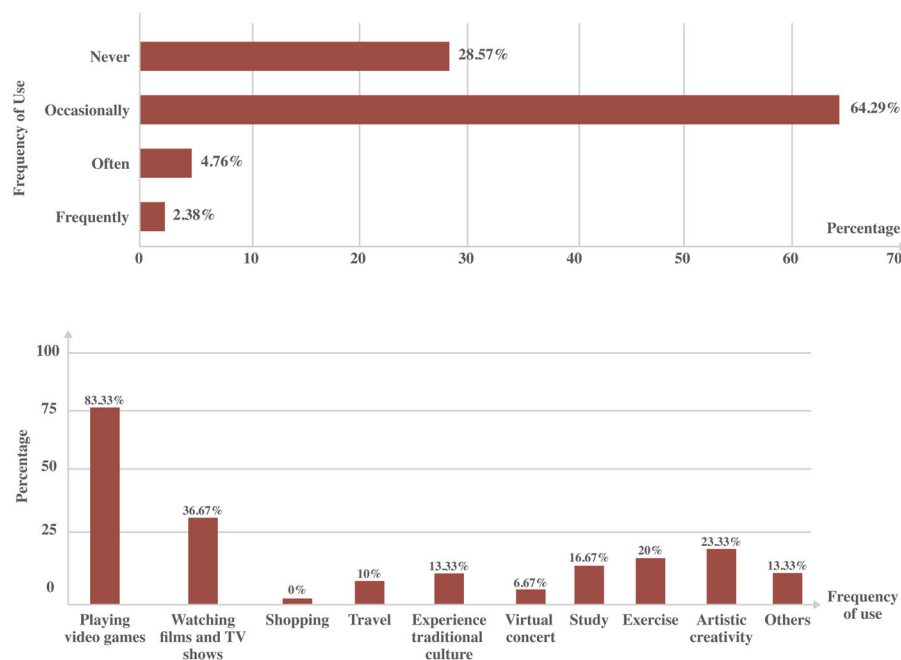


Figure 7. (a) The frequency with which participants experience VR devices (b) The activities of participants using VR devices.

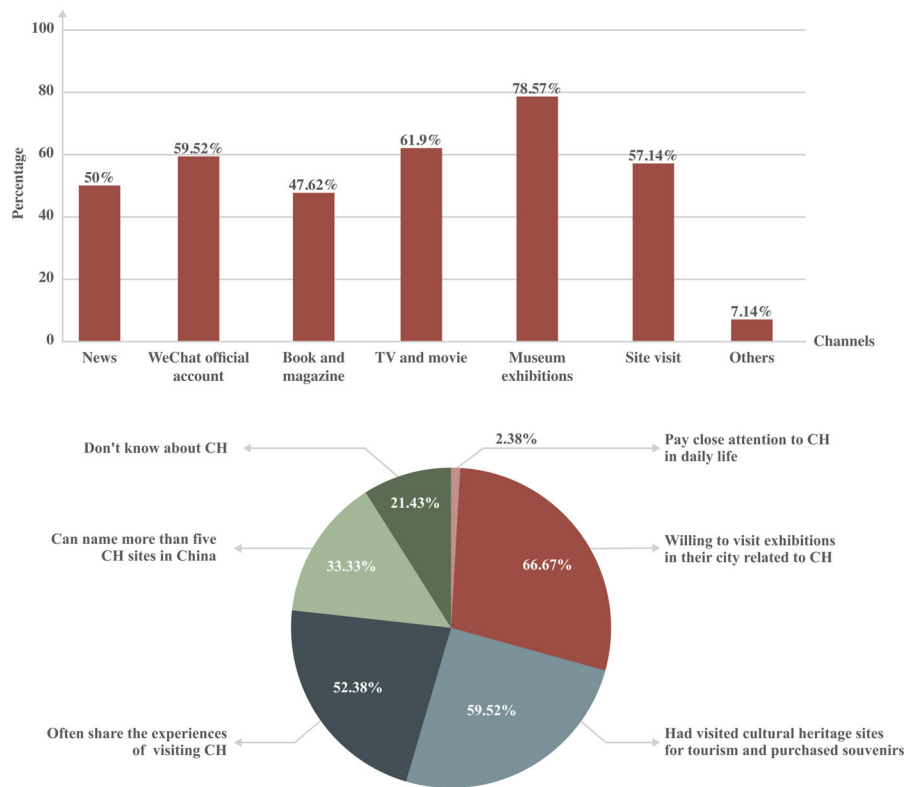


Figure 8. Survey results of: (a) various channels of acquiring CH-related knowledge (b) participants' levels of interest and attention in CH.

near-sightedness, and blurriness. Their general level of satisfaction with VR was average. However, despite the concerns about the limitations of VR experiences, all participants were still eager to experience the IVR system for CH, and we drew a similar conclusion from the interviews. In the following sections, we assessed the experiment in terms of knowledge acquisition, multimodal interaction, gamification, and storytelling.

6.1. Knowledge acquisition for CH

The majority of the participants had acquired knowledge of CH mainly through museum exhibitions (78.57%), television and films (61.90%), the official WeChat account (59.52%), on-site visits (57.14%), books (47.62%), and other channels (7.14%) (Figure 8(a)). As museum exhibitions were the main access to acquiring knowledge, our IVR system focused on the application in the museum scenario. In addition, they had also participated in CH-related exhibitions (66.67%), purchased relevant souvenirs (59.52%), and shared their experience of visiting CH sites with others (52.38%) (Figure 8(b)). Participants who wanted to experience CH-related activities were primarily interested in new technologies and experiences (35.71%) and the novelty of the activity (33.33%); alternatively, they wanted to acquire CH related knowledge (14.29%) and were motivated by curiosity (11.90%).

The results of the pre-questionnaire indicated that 40.1% of the participants were unfamiliar with the Dunhuang murals and related preservation knowledge, while only 28.6% were aware of their deterioration. Even though their

understanding of the Dunhuang murals was insufficient, 78.6% of the participants expressed a desire to learn more about the restoration experience and the technology. The majority of the participants reported being profoundly impressed by the experiment. P18 believed that, after restoring the Dunhuang mural in the IVR environment, she had a good grasp of the restoration procedure and the tools used. The experience in the IVR of Dunhuang stimulated participants' intention to further search for relevant materials on the restoration of Dunhuang murals, particularly to strengthen their understanding of the three diseases and motivate their interest in learning CH preservation knowledge after the user study. However, it is worth noting that P22 pointed out that the brief IVR experience led to a short-term memory of the expert knowledge. Specifically, she mentioned that she did not remember the names of the three diseases, but she retained the embodied knowledge of how to restore them. She said after playing the IVR, "I haven't remembered those three disease names just now, but I know how to restore them. I may go back and search online to see what they are and revisit those details."

The IVR experience also ignited the interest of Participant P4 in the Dunhuang murals. Due to the limitations imposed by the COVID-19 pandemic, she was unable to visit Dunhuang to observe the murals in person. However, she expressed her intention to proactively seek relevant knowledge through online resources. P4 recognized the potential of IVR as a medium with broad public appeal and effective information dissemination, as it allowed users to experience valuable cultural artifacts in a convenient and ecologically friendly way for CH preservation.

6.2. Establish situated consistency through the multimodal environment

6.2.1. Multimodal environments can enhance immersive and emotional experiences

There were no significant differences in the presence ($t=1.801$, $p=0.079$), engagement ($t=1.053$, $p=0.299$), usability ($t=1.22$, $p=0.23$), experience consequence ($t=1.99$, $p=0.054$), and technology adoption ($t=0.919$, $p=0.364$) dimensions of the IVE scale between the two groups. Participants of the multimodal group reported a significantly higher score compared to the comparison group in terms of immersion (Multimodal IVR: $M=8.10$, $SD=1.22$; Comparison: $M=7.14$, $SD=1.28$), flow (Multimodal IVR: $M=7.61$, $SD=1.28$; Comparison: $M=6.73$, $SD=1.30$), emotion (Multimodal IVR: $M=7.74$, $SD=1.29$; Comparison: $M=6.95$, $SD=0.98$), and skill (Multimodal IVR: $M=8.33$, $SD=1.30$; Comparison: $M=7.33$, $SD=1.23$). Participants using multimodal IVR also reported higher total scores of the IVE for their whole experiences ($M=7.68$, $SD=0.98$) in comparison with that of Comparison ($M=6.92$, $SD=0.88$) ($t=2.655$, $p=0.011$). Figure 9 depicts the mean (M) and standard deviation (SD) of the two groups. In the following section, we analyzed the impact of multimodal interaction on participants' IVR experience by examining their perceived senses, including visual, auditory, bodily (wind), tactile (ground) and olfactory sensations. We focused on the participants in the multimodal interaction experimental group to understand how different sensory inputs influenced their overall IVR experience.

During the interviews, participants in multimodal IVR group were asked to rate the degree of feeling of the above five senses from one ("very insignificant") to five ("very strong"). Visual experience ($M=4.77$, $SD=0.53$) was the primary sense perceived by participants, followed by auditory ($M=3.91$, $SD=0.75$) and bodily (wind) ($M=3.86$, $SD=0.94$) experiences, then tactile feedback from the ground ($M=2.18$, $SD=1.01$), and, lastly, olfactory experience ($M=1.23$, $SD=0.43$).

6.2.1.1. Visual. Participants expressed their affirmation and recognition of the visual aspect, highlighting that the fully immersive 3D visual design of the virtual Dunhuang scenarios played a crucial role in enhancing immersion and the overall experience. They believed that the vast desert landscapes and the accurately scaled architectural structures contributed to a greater sense of immersion and improved aesthetic enjoyment.

Initially, I felt as if I had truly arrived in the desert. Second, I feel extremely free to experience it. I am free to travel through the desert and can appreciate the scenery and it feels so real! (P13)

P17 expressed that when initially entering the IVR system, "the sunset glow and the golden desert of Dunhuang created a perfect harmony," aligning with her imagination of Dunhuang. This situated experience made her believe that she had truly arrived in Dunhuang. Despite the experiment taking place in November during the winter season, she felt as if she were in the scorching desert of Dunhuang's summer. This aligns with the situated metaphor elements we attempted to create through spatial design.

6.2.1.2. Auditory. The auditory experience was primarily reflected in the sound effects during interaction with virtual objects, task completion prompts, background music in the three scenes, and the voice assistance from the virtual character "Xiaohua." Some participants felt that the combination of background music in Chinese traditional tones made them feel calm and relaxed in the desert scene. The background music in the cave scene helped participants focus more on the task of restoring the murals. The processed 3D surround sound enhanced participants' perception of spatial location, providing a sense of presence and immersion.

Initially, I could feel the pipa music with Dunhuang characteristics, and it changed as I entered the cave to let me concentrate on restoring the murals. (P17)

Participants also reported that the voice assistance provided by "Xiaohua" could be used as a means to enhance the memory of CH knowledge input. Compared to a single

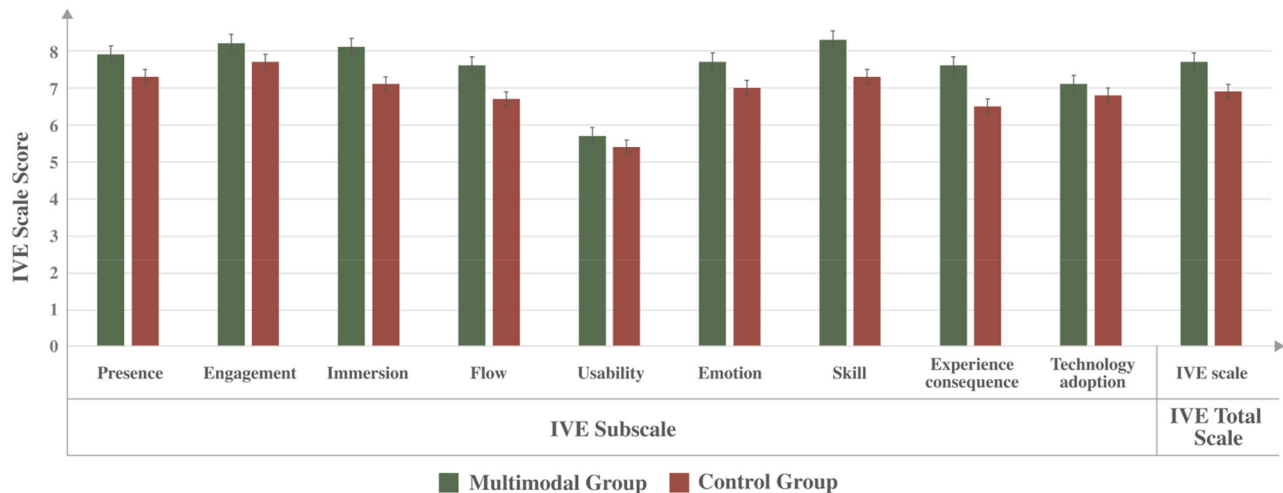


Figure 9. Mean and standard deviation of various IVE scale dimensions.

channel of vision, the audio-visual dual channel could assist people in learning more and strengthening their memories.

The voice used to introduce restorative knowledge is crucial. For this part of the learning process, auditory input is preferred over visual input. One glimpse at the learning material will be quickly forgotten, but if you listen with your ears, you must force yourself to finish before moving on to the next activity. (P13)

6.2.1.3. Bodily (wind). The wind settings created bodily experiences that brought noticeable changes in perceived temperature for the participants, enhancing the situated consistency of the virtual environment. During the experiment with P4, we observed that she exhibited increased movement frequency and engaged in more exploratory behaviors when perceiving the wind, indicating behavioral motivation. After the experience, she remarked: *"I was most impressed with the wind. It made the setting feel more authentic. I've never participated in a project that involved wind before. This time, when I gazed at the desert in front of me, there was a gust of wind. It felt as though the wind was coming from the desert itself"*

P18 emphasized the bodily experiences induced by the wind, which made him feel relaxed and triggered his desire to *"sit in the desert and play with the sand when there was a sudden wind."* This experience allowed him to form more specific and associated imaginations related to the geographical features of the CH within the IVR system. It further facilitated situated sense-making and personal interpretations of the CH environment at an individual experiential level.

The variation in perceived temperature brought about by the wind also serves as an informative cue for users during the transition between outdoor and indoor scenes. It provides a seamless transition and mitigates the discomfort caused by sudden visual scene changes. P7 expressed, *"When I entered the Nine-Storey Pagoda from the desert, it felt like the cave was genuinely chilly, and the presence of wind greatly enhanced the feeling of immersion."*

P9, who had not previously visited Dunhuang in person, reported that the experience of the wind was the most delightful surprise for him. He believed that the actual travel experience to the Mogao Grottos in Dunhuang would be consistent with the IVR experience.

Even though I don't know what the actual cave is like, the wind is fascinating. It makes the room cold as if I am really in a cave. I think visiting the Mogao Grottos would give me a similar feeling. (P9)

The findings supported our design consideration on multimodal interaction design, which brought sensory stimulation to users physical experiences, effectively enhanced situated sense-making and increased users emotional attachment to the virtual environment of CH. Therefore, when designing IVR systems for CH presentation, designers should consider connecting users physical experience and their digital experience of the virtual environment through multimodal interaction design for immersiveness.

6.2.1.4. Tactile (ground). Moving in our IVR system could be controlled by the handle or by walking in the actual

surroundings. Due to the constraints of the site, the majority of the participants preferred to control their operations through the handle, and their mobility was limited to the rotation adjustment direction. Despite the lack of ground movement, the participants still perceived "different" modes of walking.

Walking in the desert gave me a sensation of ups and downs, unlike walking on the ground. (P1)

There is a mild sensation when walking. Similar to walking in the desert. Later, when I arrived at the cave, the sensation vanished. (P8)

Although participants believed that the initial contact of their feet on the ground surface was changing and had a certain level of novelty, the impact on their spatial perception and task performance was relatively weak. P8 suggests adding sand and other fillers to the air cushion or directly depositing sand on the ground, as this approach may generate a more authentic representation of the desert.

6.2.1.5. Olfactory. The olfactory device was connected to the wind switch in order to enhance the dispersion of odors under the influence of wind. However, due to the large size of the site, as well as the participants wearing masks throughout the experiment due to the COVID-19 pandemic, their perception of odor changes was minimal. P31, who had been to Dunhuang, stated that scent was the most effective way to enhance the feeling of immersion:

Dunhuang has the unique odor of dark caves. When you close your eyes and smell that drippy odor, you will feel as if you are in the cave itself.

While the impression of Dunhuang's unique odor requires further investigation and improvement to simulate the vivid and rich smelling experience that was true to the real environment, according to the above data, the multimodal interaction settings were, in fact, favorable to the formation of situated consistency and had a significant influence in boosting the sense of immersion. However, there was no evidence to support the concept of signal hints and direction guidance that we initially proposed. It might be the fact that participants have not matched the wind and odor with the prompt directions, making it difficult to reach a consensus without guidance.

6.2.2. Task load is independent of multimodal settings

Although the multi-modal environment settings significantly improved immersion, emotion, skill, and the IVE scale, they had little effect on the task load. In terms of mental demand, physical demand, temporal demand, performance, effort, and frustration, the NASA-TLX questionnaire revealed that there was no significant difference between the two groups. Figure 10 displays the specific results. The insignificance of the experimental results might be attributable to the degree of task load that was mostly connected to the set task, but the tasks of the two groups were identical. It was quite difficult to assess the task load based simply on external multimodal elements. Second, the task complexity level

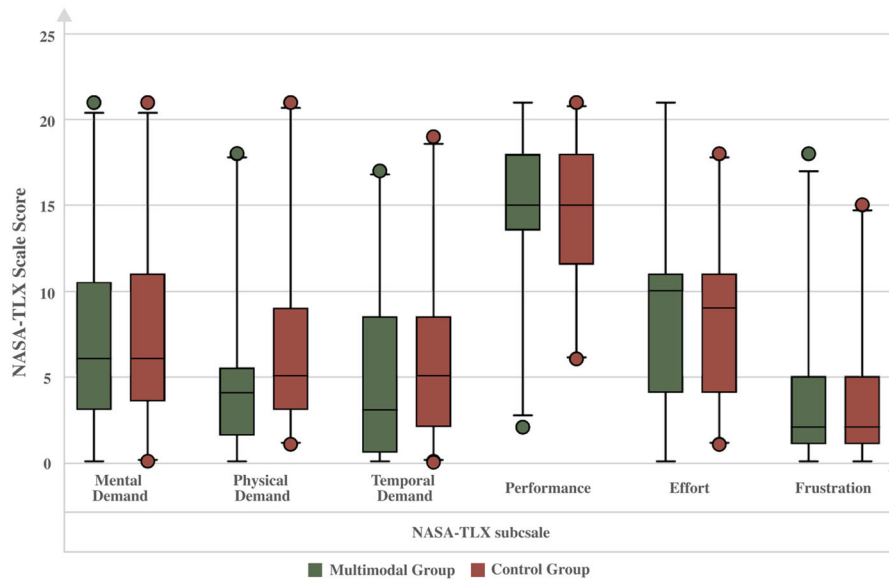


Figure 10. The median, upper, and lower quartiles of the NASA-TLX scores in the two groups.

of this IVR experience project was low, and all the participants were in an amused mood. They all experienced low mental, physical, and temporal demands and frustrations, as well as high-performance satisfaction. Therefore, a preliminary conclusion might be drawn that multimodal interaction settings have no substantial effect on task load while the task conditions remain unchanged.

6.3. Promote embodied motivation through gamification

The gamification design in the IVR system combines behavioral motivation and role motivation to facilitate the acquisition of knowledge related to the restoration of the Dunhuang murals. The primary objective of gamification is to allow users to learn about CH while enjoying fun and relaxation from interaction. As the story progresses, the increasing difficulty of tasks and the reward mechanism make it easier for users to engage with the professional knowledge of CH and maintain their attention and interest over time.

Role motivation is reflected in the identity correspondence of the users in the experience. Immersive role motivation can transform users' identities, foster empathy, and enhance learning input and knowledge creation (Ji et al., 2021). Within just 15 minutes, the perceptions of a few participants underwent a shift in their identities. They consistently considered themselves as mere visitors in the desert, at the Nine-Storey Pagoda, and in the gallery scenes. However, in the mural restoration cave, they began to feel like apprentices or junior restorers. This led to a different perception of the non-player character and a sense of trust and dependence on "Xiaohua." P18 believed that in the desert scene, Xiaohua was merely a tour guide for CH, while he saw himself as a tourist. However, in the cave, when he was asked to work independently, he thought Xiaohua became like a master restorer teaching mural restoration, and he relied heavily

on her instructions. P18 remarked, "Without Xiaohua, I might not have known what I needed to do and completed tasks so smoothly."

After successfully restoring the murals, participants experienced a deep sense of satisfaction and accomplishment, establishing an emotional connection with the cultural heritage of Dunhuang. P10 described himself as an "isolated and solitary guardian" and, upon successfully restoring the murals, felt a sense of responsibility to safeguard CH. P10 stated, "This 'guard' refers to the protection of traditional culture as well as the instinct to guard beauty." This transformation in self-perception effectively enhanced participants' understanding of the current state of CH and stimulated their awareness of the need for its preservation.

If Xiaohua played the role of a teacher and guide, the Nine-Colored Deer added a cryptic and interesting element to the digital experience, creating a relaxed and enjoyable atmosphere. P1 expressed excitement towards the Nine-Colored Deer, and expressed a desire for more interactions with it.

By designing gamified tasks, props, and environments, users' behavioral motivation is stimulated to acquire embodied knowledge. Participants expressed that bodily experiences provided them with a better understanding of the knowledge of Dunhuang mural restoration compared to video and text descriptions. P28 was particularly impressed with how the collected materials were used in the creation of mural art, said, "I was more impressed about the wisdom of the ancients in utilizing materials from nature to create such significant mural art," and expressed the desire to revisit the desert chapter to review material-related knowledge. P24, on the other hand, developed a strong curiosity about the historical stories behind the restored murals and wished to further explore the uses of restoration tools for other diseases:

The corridor introduced the murals of Dunhuang, which depict the relationship between gods and humans, the appearance of



Figure 11. Users' paintings from the free creation session.

gods, and their legend. After acquiring the tools, I wanted to know what kind of gods and legends were depicted in the paintings I would restore in the future. How can these tools be used for other diseases? I was eager to restore more murals and get a deeper understanding of their stories.

After the restoration was complete, users were able to paint freely in the creative phase. Although the configuration at this stage was preliminary and the precision and color of the brush were insufficient, they were nevertheless highly interested in this high degree of freedom and flexible interaction mode, and they also attempted to create their own masterpieces (Figure 11).

6.4. Create affective connectedness between users and CH through storytelling

In this study, storytelling facilitated affective connectivity and knowledge transfer. The narrative embodiment, with characters like “Xiaohua” and the Nine-Colored Deer, greatly assisted users. Participants perceived the narrative approach as narrowing the gap between themselves and CH through emotional connection. Within the immersive IVR environment, users could closely examine the Dunhuang murals without worrying about damage from carbon dioxide pollution. This reduced the perceived distances in space,

time, and emotional connection between users and the murals.

I have never been to Dunhuang, and now that I understand it, I would not dare to visit. I am concerned that the CO₂ that I breathe will cause extra deterioration. (P28)

When I visited Dunhuang, I went to view several caves, and the staff urged me to move quickly because they worried that carbon dioxide from tourists' breath would deteriorate the murals. I waited in line for approximately two hours but only spent five seconds inside. (P31)

I think VR is incredibly helpful in understanding CH. It may transcend time and space. Even if I have never visited Dunhuang, I can still appreciate the caves. I can closely observe the facial emotions and clothing on the murals. This kind of experience cannot be had in an actual cave. (P17)

The IVR system had a significant impact on transforming the participants' mindset. Before the experiments, the majority of our participants viewed the Dunhuang Mogao Grottoes as mere spectators and were unaware of their value. After the experience, however, they began to comprehend the severity of the mural's deterioration and the necessity for restoration. P13 expressed his feelings of regret and determination for CH preservation: *“The murals of Dunhuang are part of the CH. Once they vanish, they will vanish completely. I feel deep regret. The deteriorated mural has been*

eliminated. We must conserve the mural and prevent its deterioration."

After the experience, P14 not only gained knowledge about the Dunhuang murals but also formed opinions on mural restoration. He believed that mural restoration required a combination of the restorer's expertise and intuition, making it, to some extent, as valuable as the creation of the murals themselves. The storytelling approach in the IVR system facilitated a gradual emotional transformation, shifting users from an objective perspective of cultural heritage to developing empathy and appreciating the beauty of cultural heritage on a subjective level. This emotional connection also fostered an attachment to the cultural heritage sites, thereby stimulating their awareness and commitment to CH preservation.

7. Conclusion

Most CH sites face challenges in achieving permanent physical presentation for tourism purposes. However, digital technology offers the opportunity for users to experience CH virtually, transcending the limitations of time and space, especially in the post-pandemic era. The immersive situated experience enabled by IVR systems has become an important approach to fostering interest in CH, disseminating knowledge, and promoting preservation awareness. While existing research on IVR systems for CH has primarily focused on audiovisual experiences and 3D recurrence of CH sites, there is limited exploration and discussion of the development of user-centered digital experiences of IVR systems for CH that incorporate a multimodal environment, embodied interaction, and emotional engagement.

Therefore, in response to the three research questions, we first positioned the design requirements of IVR systems to support CH digital experiences by conducting an extensive review of relevant literature. These requirements encompassed multimodal interaction, gamification, and storytelling. Secondly, based on the identified design requirements, we proposed a three-phase design strategy to enhance the user-centered experience of IVR systems for CH presentation, development, and promotion through three main design considerations: situated consistency, embodied motivation, and affective connectedness. Subsequently, we applied this strategy to design an IVR system to engage users in experiencing the restoration process of the Mogao Murals, providing a digital tour of Dunhuang that encompasses scenarios ranging from preparing raw materials in the desert, accumulating knowledge of restoration techniques, to repairing murals within the cave.

To evaluate the effectiveness of our IVR system, we conducted a between-subject user study involving 42 participants, and employed a combination of qualitative and quantitative methods for analysis. The results demonstrated that the IVR system, designed in accordance with our strategy, significantly improved the users' VR immersive and emotional experience, facilitating a more systematic understanding of the local historical, natural, and social environmental information associated with the CH site. Moreover, this design strategy can be

applied not only to the Dunhuang Mogao Grottoes but also to other CH experiences within museum contexts for knowledge dissemination to the general public. This study also hoped to empower multiple stakeholders involved in CH promotion, including managers, researchers, designers, and developers, by fostering a more equitable and inclusive approach rooted in user-centered design. By actively engaging individuals in the experience, our work aims to bridge the knowledge gap between experts and the general public while promoting the dissemination of authoritative expert knowledge in a comprehensible manner.

8. Limitations and future work

Our research has demonstrated that IVR systems can effectively motivate users' interests and facilitate their active learning of CH preservation knowledge. However, based on user feedback, we acknowledge that our system needs improvement in terms of the continuity of the plot, the naturalness of the interactions, and the degree of multimodality. CH sites such as Dunhuang Mogao Grottoes possess complex dimensions intertwined with local geographical and historical factors. Therefore, more intricate and captivating designs are necessary for both large-scale explorations to increase immersiveness and tightly focused interventions to improve users' retention. Some of the somatosensory experiences derived from multimodal interaction did not align seamlessly with the VR imagery. For instance, in the desert scenario, the environment should exhibit more dynamic elements when fans create the sensation of wind. During transitions from outdoor to indoor environments, adjustments in smell concentration and body temperature could be made to provide a smoother and more natural experience. We also found that users had significant interest in and enthusiasm for user-generated content in the final free creation session, despite the limited number of customization choices. These interactions should be optimized and iterated with further review and consideration based on our results.

The CH-oriented IVR system designed and presented in this article is a preliminary exploration of the proposed design strategy. In the future, we plan to enhance the mapping relations between digital and physical experiences to further create situated consistency and stimulate embodied motivation. We expect to continue exploring how to create access for the general public to develop a deeper understanding of and build affective connectedness with CH in a relatively short period of time. We wish to focus more on participatory design and the social attributes of the CH-oriented IVR system and further explore the intersection of social computing and CH-based IVR systems to stimulate the public's interest and broaden the impact of knowledge dissemination on a social level.

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References

- Antle, K., Bykersma, M., Mortimer, M., Vickers-Rich, P., Rich, T., Horan, B. (2018). *Real-world data for virtual reality experiences: Interpreting excavations* [Paper presentation]. 2018 3rd Digital Heritage International Congress Held Jointly with 2018 24th International Conference on Virtual Systems & Multimedia (pp. 1–8). <https://doi.org/10.1109/DigitalHeritage.2018.8810060>
- Beck, J., Rainoldi, M., & Egger, R. (2019). Virtual reality in tourism: A state-of-the-art review. *Tourism Review*, 74(3), 586–612. <https://doi.org/10.1108/TR-03-2017-0049>
- Bekele, M. K., Pierdicca, R., Frontoni, E., Malinverni, E. S., & Gain, J. (2018). A survey of augmented, virtual, and mixed reality for cultural heritage. *Journal on Computing and Cultural Heritage*, 11(2), 1–36. <https://doi.org/10.1145/3145534>
- Brancati, N., Caggianese, G., Frucci, M., Gallo, L., & Neroni, P. (2017). Experiencing touchless interaction with augmented content on wearable head-mounted displays in cultural heritage applications. *Personal and Ubiquitous Computing*, 21(2), 203–217. <https://doi.org/10.1007/s00779-016-0987-8>
- Ceccacci, S., Generosi, A., Leopardi, A., Mengoni, M., & Mandorli, F. (2021). The role of haptic feedback and gamification in virtual museum systems. *Journal on Computing and Cultural Heritage*, 14(3), 1–14. <https://doi.org/10.1145/3453074>
- Cecotti, H. (2022). Cultural heritage in fully immersive virtual reality. *Virtual Worlds*, 1(1), 82–102. <https://doi.org/10.3390/virtualworlds1010006>
- Chrysanthakopoulou, A., Kalatzis, K., & Moustakas, K. (2021). Immersive virtual reality experience of historical events using haptics and locomotion simulation. *Applied Sciences*, 11(24), 11613. <https://doi.org/10.3390/app112411613>
- Chu, J. H., & Mazalek, A. (2019). Embodied engagement with narrative: A design framework for presenting cultural heritage artifacts. *Multimodal Technologies and Interaction*, 3(1), 1. <https://doi.org/10.3390/mti3010001>
- de Paiva Guimarães, M., Martins, J. M., Dias, D. R. C., Guimarães, R. D. F. R., & Gnecco, B. B. (2022). An olfactory display for virtual reality glasses. *Multimedia Systems*, 28(5), 1573–1583. <https://doi.org/10.1007/s00530-022-00908-8>
- Dima, M., Hurcombe, L., & Wright, M. (2014). Touching the past: Haptic augmented reality for museum artefacts. In R. Shumaker, & S. Lackey (Eds.), *Virtual, augmented and mixed reality: Applications of virtual and augmented reality* (pp. 3–14). Springer. https://doi.org/10.1007/978-3-319-07464-1_1
- Dincelli, E., & Yayla, A. (2022). Immersive virtual reality in the age of the Metaverse: A hybrid-narrative review based on the technology affordance perspective. *The Journal of Strategic Information Systems*, 31(2), 101717. <https://doi.org/10.1016/j.jsis.2022.101717>
- Ferdani, D., & Pietroni, E. (2021). Virtual restoration and virtual reconstruction in cultural heritage: Terminology, methodologies, visual representation techniques and cognitive models. *Information*, 12(4), 167. <https://doi.org/10.3390/info12040167>
- Fu, X., Zhu, Y., Xiao, Z., Xu, Y., & Ma, X. (2020). RestoreVR: Generating embodied knowledge and situated experience of Dunhuang mural conservation via interactive virtual reality. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (pp. 1–13). <https://doi.org/10.1145/3313831.3376673>
- Fu, X. Y., Han, Y., Sun, Z. J., Ma, X. J., & Xu, Y. Q. (2017). Line-drawing enhanced interactive mural restoration for Dunhuang Mogao Grottoes. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, IV-2/W2, 99–106. <https://doi.org/10.5194/isprs-annals-IV-2-W2-99-2017>
- Kenderdine, S. (2015). Embodiment, entanglement, and immersion in digital cultural heritage. *A new companion to digital humanities*, 22–41. <https://doi.org/10.1002/9781118680605.ch2>
- Galatis, P., Gavalas, D., Kasapakis, V., Pantziou, G. E., & Zaroliagis, C. D. (2016). Mobile augmented reality guides in cultural heritage. In *MobiCASE* (pp. 11–19). <https://doi.org/10.4108/eai.30-11-2016.2266954>
- Gibbs, R. W., Jr., Lima, P. L. C., & Francozo, E. (2004). Metaphor is grounded in embodied experience. *Journal of Pragmatics*, 36(7), 1189–1210. <https://doi.org/10.1016/j.pragma.2003.10.009>
- Guidi, G., Frischer, B., Lucenti, I. (2007). *Rome reborn-virtualizing the ancient imperial Rome* [Paper presentation]. Workshop on 3D Virtual Reconstruction and Visualization of Complex Architectures. https://www.isprs.org/proceedings/XXXVI/5-W47/pdf/guidi_et_al.pdf
- Han, P. H., Chen, Y. S., Lee, K. C., Wang, H. C., Hsieh, C. E., Hsiao, J. C., Chou, C. H., & Hung, Y. P. (2018). Haptic around: Multiple tactile sensations for immersive environment and interaction in virtual reality. In *Proceedings of the 24th ACM Symposium on Virtual Reality Software and Technology* (pp. 1–10). <https://doi.org/10.1145/3281505.3281507>
- Han, P., Chen, Y., Liu, I., Jang, Y., Tsai, L., Chang, A., & Hung, Y. (2020). A compelling virtual tour of the Dunhuang Cave with an immersive head-mounted display. *IEEE Computer Graphics and Applications*, 40(1), 40–55. <https://doi.org/10.1109/MCG.2019.2936753>
- Hart, S. G., & Staveland, L. E. (1988). Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research. In *Advances in psychology* (Vol. 52, pp. 139–183). [https://doi.org/10.1016/S0166-4115\(08\)62386-9](https://doi.org/10.1016/S0166-4115(08)62386-9)
- Hay, K. E., & Barab, S. A. (2001). Constructivism in practice: A comparison and contrast of apprenticeship and constructionist learning environments. *Journal of the Learning Sciences*, 10(3), 281–322. https://doi.org/10.1207/S15327809JLS1003_3
- Hayward, V., Astley, O. R., Cruz-Hernandez, M., Grant, D., & Robles-De-La-Torre, G. (2004). Haptic interfaces and devices. *Sensor Review*, 24(1), 16–29. <https://doi.org/10.1108/02602280410515770>
- Herrera, N. S., & McMahan, R. P. (2014). Development of a simple and lowcost olfactory display for immersive media experiences. In *Proceedings of the 2nd ACM International Workshop on Immersive Media Experiences*, 1–6. <https://doi.org/10.1145/2660579.2660584>
- Hu-Au, E., & Lee, J. J. (2017). Virtual reality in education: A tool for learning in the experience age. *International Journal of Innovation in Education*, 4(4), 215–226. <https://doi.org/10.1504/IJIE.2017.091481>
- Ibrahim, N., & Ali, N. M. (2018). A conceptual framework for designing virtual heritage environment for cultural learning. *Journal on Computing and Cultural Heritage*, 11(2), 1–27. <https://doi.org/10.1145/3117801>
- Ji, Y., Zhong, S., Zhang, B., & Sean, C. (2021). Constructing embodied interaction of intangible cultural heritage course through immersive virtual reality. In *The Ninth International Symposium of Chinese CHI*, 133–138. <https://doi.org/10.1145/3490355.3490373>
- Jin, L., Bo, L. W., Song, L. L., Wu, J., & Yu, T. X. (2021). The design and implementation of virtual reality display systems in Mogao Cave 249. *Dunhuang Research*, 188(4), 133–141. <https://doi.org/10.13584/j.cnki.issn1000-4106.2021.04.016>
- Kobayashi, M., Kon, Y., & Kajimoto, H. (2018). Presentation of stepping up and down by pneumatic balloon shoes device. In H. Kajimoto, D. Lee, S.-Y. Kim, M. Konyo, & K.-U. Kyung (Eds.), *Haptic interaction: Perception, devices and algorithms* (pp. 317–320). Springer. https://doi.org/10.1007/978-981-13-3194-7_69
- Komerska, R., & Ware, C. (2003). Haptic task constraints for 3D interaction. In *11th Symposium on Haptic Interfaces for Virtual*

- Environment and Teleoperator Systems*, 270–277. <https://doi.org/10.1109/HAPTIC.2003.1191295>
- Konstantakis, M., Alexandridis, G., & Caridakis, G. (2020). A personalized heritage-oriented recommender system based on extended cultural tourist typologies. *Big Data and Cognitive Computing*, 4(2), 12. <https://doi.org/10.3390/bdcc4020012>
- Konstantakis, M., & Caridakis, G. (2020). Adding culture to UX: UX research methodologies and applications in cultural heritage. *Journal on Computing and Cultural Heritage*, 13(1), 1–17. <https://doi.org/10.1145/3354002>
- Konstantakis, M., Christodoulou, Y., Aliprantis, J., & Caridakis, G. (2022). ACUX Recommender: A Mobile Recommendation System for Multi-Profile Cultural Visitors Based on Visiting Preferences Classification. *Big Data and Cognitive Computing*, 6(4), 144. <https://doi.org/10.3390/bdcc6040144>
- Konstantakis, M., Kalatha, E., & Caridakis, G. (2019). Cultural heritage, serious games and user personas based on Gardner's theory of multiple intelligences: "The Stolen Painting" game. *International Conference on Games and Learning Alliance*, 11899, 490–500. https://doi.org/10.1007/978-3-030-34350-7_47
- Lécuyer, A., Mobuchon, P., Mégard, C., Perret, J., Andriot, C., & Colinet, J. P. (2003). HOMERE: A multimodal system for visually impaired people to explore virtual environments. In *Proceedings of IEEE Virtual Reality*, 251–258. <https://doi.org/10.1109/VR.2003.1191147>
- Lee, T. H., & Shen, Y. L. (2013). The influence of leisure involvement and place attachment on destination loyalty: Evidence from recreationists walking their dogs in urban parks. *Journal of Environmental Psychology*, 33, 76–85. <https://doi.org/10.1016/j.jenvp.2012.11.002>
- Leow, F.-T., & Ch'ng, E. (2021). Analysing narrative engagement with immersive environments: Designing audience-centric experiences for cultural heritage learning. *Museum Management and Curatorship*, 36(4), 342–361. <https://doi.org/10.1080/09647775.2021.1914136>
- Li, P., Pan, B., Pan, H. (2019). Application of VR technology in cultural heritage dissemination: Design and production of VR documentary "Dunhuang". *International Conference on Machinery, Mechanics, Materials, and Computer Engineering*, 133–143. <https://doi.org/10.25236/mmmce.2019.031>
- Li, M., & Xu, Y. Q. (2021). Research on information design from the perspective of tangible interactive narratives. *Zhuangshi*, 64(9), 24–28. <https://doi.org/10.16272/j.cnki.cn11-1392/j.2021.09.001>
- Li, M., Wang, Y., & Xu, Y. Q. (2022). Computing for Chinese cultural heritage. *Visual Informatics*, 6(1), 1–13. <https://doi.org/10.1016/j.vis-inf.2021.12.006>
- Liarokapis, F., Petridis, P., Andrews, D., & de Freitas, S. (2017). Multimodal serious games technologies for cultural heritage. In M. Ioannides, N. Magnenat-Thalmann, & G. Papagiannakis (Eds.), *Mixed reality and gamification for cultural heritage* (pp. 371–392). Springer. https://doi.org/10.1007/978-3-319-49607-8_15
- Lindeman, R. W., & Yanagida, Y. (2003). Empirical studies for effective near-field haptics in virtual environments. In *Proceedings of IEEE Virtual Reality*, 287–288. <https://doi.org/10.1109/VR.2003.1191166>
- Low, S. M., & Altman, I. (1992). Place attachment. In S. M. Low, & I. Altman (Eds.), *Place attachment* (pp. 1–12). Springer. https://doi.org/10.1007/978-1-4684-8753-4_1
- Ma, X. N., Tu, L., & Xu, Y. Q. (2019). Development status of the digitization of intangible cultural heritages. *Scientia Sinica Informationis*, 49(2), 121–142. <https://doi.org/10.1360/N112018-00201>
- Mariotti, S. (2022). Gamifying cultural heritage. Education, tourism development, and territory promotion: Two Italian examples. In O. Bernardes, V. Amorim, & A. C. Moreira (Eds.), *Handbook of research on cross-disciplinary uses of gamification in organizations* (pp. 418–444). IGI Global. <https://doi.org/10.4018/978-1-7998-9223-6.ch020>
- Micoli, L., Barsanti, S. G., Malik, U., & Guidi, G. (2018). 3D data integration for the digital reconstruction of cultural heritage monuments. *IOP Conference Series: Materials Science and Engineering*, 364(1), 012043. <https://doi.org/10.1088/1757-899X/364/1/012043>
- Nakaizumi, F., Noma, H., Hosaka, K., & Yanagida, Y. (2006). SpotScents: A novel method of natural scent delivery using multiple scent projectors. In *IEEE Virtual Reality Conference*, 207–214. <https://doi.org/10.1109/VR.2006.122>
- Pan, Z. G., Yuan, Q. S., Chen, S. N., & Zhang, M. M. (2020). State of the art on the digital presentation and interaction of culture heritage. *Zhejiang Daxue Xuebao (Lixue Ban)* 47(3), 261–273. <https://doi.org/10.3785/j.issn.1008-9497.2020.03.001>
- Pietroni, E., & Adami, A. (2014). Interacting with virtual reconstructions in museums: The etruscanning project. *Journal on Computing and Cultural Heritage*, 7(2), 1–29. <https://doi.org/10.1145/2611375>
- Plecher, D. A., Wandinger, M., & Klinker, G. (2019). Mixed reality for cultural heritage. Proceedings of the. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, 1618–1622. <https://doi.org/10.1109/VR.2019.8797846>
- Ranasinghe, N., Jain, P., Thi Ngoc Tram, N., Koh, K. C. R., Tolley, D., Karwita, S., Lien-Ya, L., Liangkun, Y., Shamaiah, K., Eason Wai Tung, C., Yen, C. C., & Do, E. Y. L. (2018). Season traveller: Multisensory narration for enhancing the virtual reality experience. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, Paper 577, 1–13. <https://doi.org/10.1145/3173574.3174151>
- Reinhard, C. D., & Dervin, B. (2012). Comparing situated sense-making processes in virtual worlds: Application of Dervins Sense-Making Methodology to media reception situations. *The International Journal of Research into New Media Technologies*, 18(1), 27–48. <https://doi.org/10.1177/1354856511419914>
- Ren, W., & Chen, X. (2021). Evaluation of the online virtual reality 360° world cultural heritage tourism under the charter on interpretation and presentation of cultural heritage during the Covid-2019 outbreak. *Open Archaeology*, 7(1), 1192–1215. <https://doi.org/10.1515/opar-2020-0188>
- Sallnäs, E. L., Rasmus-Gröhn, K., & Sjöström, C. (2000). Supporting presence in collaborative environments by haptic force feedback. *ACM Transactions on Computer-Human Interaction*, 7(4), 461–476. <https://doi.org/10.1145/365058.365086>
- Schmidt, D., Kovacs, R., Mehta, V., Umaphathi, U., Köhler, S., Cheng, L. P., & Baudisch, P. (2015). Level-ups: Motorized stilts that simulate stair steps in virtual reality. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, 2157–2160. <https://doi.org/10.1145/2702613.2725431>
- Simon, N., William, F., Peter, L., Marie, E., & Jonas, K. O. (2023). A graspable olfactory display for virtual reality. *International Journal of Human-Computer Studies*, 169, 102928. <https://doi.org/10.1016/j.ijhcs.2022.102928>
- Škola, F., Rizvić, S., Cozza, M., Barbieri, L., Bruno, F., Skarlatos, D., & Liarokapis, F. (2020). Virtual reality with 360-video storytelling in cultural heritage: Study of presence, engagement, and immersion. *Sensors*, 20(20), 5851. <https://doi.org/10.3390/s20205851>
- Slater, M., & Sanchez-Vives, M. V. (2016). Enhancing our lives with immersive virtual reality. *Frontiers in Robotics and AI*, 3(73), 1–47. <https://doi.org/10.3389/frobt.2016.00074>
- Tcha-Tokey, K., Christmann, O., Loup-Escande, E., & Richir, S. (2016). Proposition and validation of a questionnaire to measure the user experience in immersive virtual environments. *International Journal of Virtual Reality*, 16(1), 33–48. <https://doi.org/10.20870/IJVR.2016.16.1.2880>
- Tcha-Tokey, K., Loup-Escande, E., Christmann, O., & Richir, S. (2017). Effects on user experience in an edutainment virtual environment: Comparison between CAVE and HMD. In *Proceedings of the European conference on cognitive ergonomics*, 1–8. <https://doi.org/10.1145/3121283.3121284>
- Teng, S. Y., Lin, C. L., Chiang, C. H., Kuo, T. S., Chan, L., Huang, D. Y., & Chen, B. Y. (2019). TilePoP: Tile-type pop-up prop for virtual reality. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology*, 639–649. <https://doi.org/10.1145/3332165.3347958>
- Themistocleous, K., Cuca, B., Agapiou, A., Lysandrou, V., Tzouvaras, M., Hadjimitsis, D. G., Kyriakides, O., Kouhartsiouk, D., Margottini, C., Spizzichino, D., Cigna, F., Crosta, G., Frattini, P., & Merodo, J. A. F. (2016). The protection of cultural heritage sites from geohazards: The PROTHEGO project. In *Digital Heritage. Progress in Cultural Heritage: Documentation, Preservation, and Protection*, EuroMed 2016. Lecture Notes in Computer Science (Vol. 10059, pp. 91–98). Springer. https://doi.org/10.1007/978-3-319-48974-2_11

- Trauer, B., & Ryan, C. (2005). Destination image, romance and place experience – An application of intimacy theory in tourism. *Tourism Management*, 26(4), 481–491. <https://doi.org/10.1016/j.tourman.2004.02.014>
- Trichopoulos, G., Aliprantis, J., Konstantakis, M., Michalakis, K., & Caridakis, G. (2022). Tangible and personalized DS application approach in cultural heritage: The CHATS Project. *Computers*, 11(2), 19. <https://doi.org/10.3390/computers11020019>
- van den Eede, Y. (2015). Tracing the tracker: A postphenomenological inquiry into self-tracking technologies. In R. Rosenberger, & P-P. Verbeek (Eds.), *Postphenomenological investigations: Essays on human-technology relations* (pp. 143–158). Lexington Books.
- Watkins, C. J. (2002). *Methods and apparatus for localized delivery of scented aerosols* (U.S. Patent No. 6,357,726). U.S. Patent and Trademark Office.
- Wu, Q. (2019). Digital experience design for cultural heritage. *Zhuangshi*, 62(1), 12–15. <https://doi.org/10.16272/j.cnki.cn11-1392/j.2019.01.005>
- Xia, G. F., Hu, C. M., & Fan, L. (2018). A method for reconstructing precise 3D models for statues. *Dunhuang Research*, 38(3), 131–140. <https://doi.org/10.13584/j.cnki.issn1000-4106.2018.03.017>

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Appendix A. Participant self-report

Categories	Items	Description	Percentage
Demographics	Basic Information	Gender	Male
			Female
		Age	18 ~ 25
			26 ~ 30
			31 ~ 40
		Occupation	Full-time students
			Sales staff
			Administrative/Logistic Staff
			Technical/R&D personnel
		Education	High School
Self-report	VR experience		Undergraduate
			Graduate
			Doctor and above
		Familiarity with VR devices	Never used
			Occasionally used
			Often used
		Types of VR devices experience	Frequently used
			Play games
			Watch movies and videos
			Travel
			Experience traditional culture
			VR virtual concert
			Taking classes
			Sports training
			Art creation
			Other
		Places to experience VR devices	Personal use at home
			Experience in malls
			In museums
			In art exhibitions
		Whether to experience cultural heritage related VR projects in museums	Yes, I have experienced it
			No, I haven't experienced it
			Presentation and narration of cultural relics
			Cultural heritage interaction and experience category
		Problems encountered during the experience	Unable to determine their spatial location, lost in the VR environment

(continued)

Continued.

Categories	Items	Description	Percentage
Knowledge of cultural heritage	Level of satisfaction	Don't know how to take the next step, can't get effective clues	85.71%
		Feeling dizzy and physically uncomfortable	14.29%
		Nearsightedness is blurred when you take off your glasses	14.29%
		Handle positioning recognition error, resulting in frequent correction	14.29%
	Level of willingness to experience VR equipment	Quite satisfied	28.57%
		Fair	71.43%
		Very willing	73.68%
		More willing	21.05%
	Participants' levels of interest and attention in cultural heritage	Fairly	5.26%
		Pay close attention to cultural heritage in daily life	2.38%
		Willing to visit exhibitions in their city related to cultural heritage	66.67%
		Had visited cultural heritage sites for tourism and purchased souvenirs	59.52%
	Sources of information on cultural heritage	Often share the experiences of visiting cultural heritage	52.38%
		Can name more than five cultural heritage sites in China	33.33%
		Don't know about cultural heritage	21.43%
		Mobile News	50.00%
	Expected features of cultural heritage related projects	WeChat official account	59.52%
		Books and magazines	47.62%
		TV and movies	61.90%
		Museums and exhibitions	78.57%
NASA-TLX Scales	Level of knowledge of Dunhuang murals	Site visits to cultural heritage	57.14%
		Others (introduced by friends, documentaries, contact during work)	7.14%
		Desire to acquire knowledge related to cultural heritage	14.29%
		Novelty and attractiveness of the project	33.33%
	Level of willingness to learn about Dunhuang frescoes and restoration	Want to raise their awareness of cultural heritage preservation	2.38%
		Want to experience new technologies and experiences	35.71%
		Driven by curiosity	11.9%
		Other	2.38%
	Items	Only heard of the Digital Dunhuang website	33.33%
		Visited the Digital Dunhuang website	19.05%
		Visited Mogao Grottoes at Dunhuang	9.52%
		Knew about the damage to Dunhuang murals	28.57%
	Groups	Know the methods and techniques used to restore Dunhuang murals	14.29%
		I don't know much about Dunhuang murals	40.48%
		I want to learn new experiences and techniques	78.57%
		It doesn't matter, it's okay to learn or not	21.43%
	Scores		
	Mental Demand	Multimodal groups	15(5.5 ~ 17.8)
		Control group	17.4(9 ~ 20.7)
	Physical Demand	Multimodal groups	4(1.5 ~ 5.5)
		Control group	5(3 ~ 9)
	Temporal Demand	Multimodal groups	3(0.5 ~ 8.5)
		Control group	5(2 ~ 8.5)
	Performance	Multimodal groups	15(13.5 ~ 18)
		Control group	15(11.5 ~ 18)
	Effort	Multimodal groups	10(4 ~ 11)
		Control group	9(4 ~ 11)
	Frustration	Multimodal groups	2(1 ~ 5)
		Control group	2(1 ~ 5)